

A Kv2 inhibitor combination reveals native neuronal conductances consistent with Kv2/KvS heteromers

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eLife Assessment

Some delayed rectifier currents in neurons are formed by the combination of Kv2 and silent subunits, KvS. However, we lack the tools to identify these heteromeric channels in vivo. In this **important** study by the Sack group, the authors identify a pharmacological tool that can reveal the presence of KvS subunits as components of the delayed rectifier potassium currents in selected neurons. The experimental evidence presented in the manuscript is **compelling** and represents a significant advance that should be of interest to a wide community of neuroscientists and channel physiologists.

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Abstract

KvS proteins are voltage-gated potassium channel subunits that form functional channels when assembled into heteromers with Kv2.1 (*KCNB1*) or Kv2.2 (*KCNB2*). Mammals have 10 KvS subunits: Kv5.1 (*KCNF1*), Kv6.1 (*KCNG1*), Kv6.2 (*KCNG2*), Kv6.3 (*KCNG3*), Kv6.4 (*KCNG4*), Kv8.1 (*KCNV1*), Kv8.2 (*KCNV2*), Kv9.1 (*KCNS1*), Kv9.2 (*KCNS2*), and Kv9.3 (*KCNS3*). Electrically excitable cells broadly express channels containing Kv2 subunits and most neurons have substantial Kv2 conductance. However, whether KvS subunits contribute to these conductances has not been clear, leaving the physiological roles of KvS subunits poorly understood. Here, we identify that two potent Kv2 inhibitors, used in combination, can distinguish conductances of Kv2/KvS heteromers and Kv2-only channels. We find that Kv5, Kv6, Kv8, or Kv9-containing channels are resistant to the Kv2-selective pore-blocker RY785 yet remain sensitive to the Kv2-selective voltage sensor modulator guangxitoxin-1E (GxTX). Using these inhibitors in mouse superior cervical ganglion neurons, we find predominantly RY785-sensitive conductances consistent with channels composed entirely of Kv2 subunits. In contrast, RY785-resistant but GxTX-sensitive conductances consistent with Kv2/KvS

heteromeric channels predominate in mouse and human dorsal root ganglion neurons. These results establish an approach to pharmacologically distinguish conductances of Kv2/KvS heteromers from Kv2-only channels, enabling investigation of the physiological roles of endogenous KvS subunits. These findings suggest that drugs which distinguish KvS subunits could modulate electrical activity of subsets of Kv2-expressing cell types.

Introduction

The Kv2 voltage-gated K⁺ channel subunits, Kv2.1 and Kv2.2, are broadly expressed in electrically excitable cells throughout the body and have important ion-conducting and non-conducting functions (Trimmer, 1993 [↗](#); Du et al., 2000 [↗](#); Li et al., 2013 [↗](#); Liu and Bean, 2014 [↗](#); Bishop et al., 2015 [↗](#); Johnson et al., 2018 [↗](#); Kirmiz et al., 2018 [↗](#); Vierra et al., 2021 [↗](#); Matsumoto et al., 2023 [↗](#)). Consistent with this widespread expression, Kv2 channels have profound impacts on many aspects of our physiology including vision, seizure suppression, stroke recovery, pain signaling, blood pressure, insulin secretion, and reproduction (Bocksteins, 2016 [↗](#)). Although modulation of Kv2 channels may hold therapeutic promise, Kv2 subunits are poor systemic drug targets due to their importance in many tissues.

A potential source of molecular diversity for Kv2 channels are a group of Kv2-related proteins which have been referred to as regulatory, silent, or KvS subunits (Bocksteins et al., 2009 [↗](#); Kobertz, 2018 [↗](#)). KvS subunits are an understudied class of voltage-gated K⁺ channel (Kv) subunits that comprise one fourth of mammalian Kv subunit types. Like all other Kv proteins, the ten KvS proteins (Kv5.1, Kv6.1-6.4, Kv8.1-8.2, and Kv9.1-9.3) are alpha subunits with a voltage sensor and pore domain. Distinct from other Kv alpha subunits, KvS have not been found to form functional homomeric channels. Rather, KvS alpha subunits co-assemble with Kv2 alpha subunits to form heterotetrameric Kv2/KvS channels in which the KvS subunit makes up part of the K⁺-conductive pathway (Salinas et al., 1997b [↗](#); Kramer et al., 1998 [↗](#)). Kv2/KvS heteromers have biophysical properties distinct from those of homomeric Kv2 channels (Post et al., 1996 [↗](#); Salinas et al., 1997a [↗](#); Kramer et al., 1998 [↗](#); Richardson and Kaczmarek, 2000 [↗](#); Zhong et al., 2010 [↗](#); Bocksteins et al., 2012 [↗](#); Bocksteins et al., 2017 [↗](#)). KvS mRNAs are expressed in tissue and cell-specific manners that overlap with Kv2.1 or Kv2.2 expression (Castellano et al., 1997 [↗](#); Salinas et al., 1997a [↗](#); Kramer et al., 1998 [↗](#); Bocksteins et al., 2012 [↗](#); Bocksteins and Snyders, 2012 [↗](#); Bocksteins, 2016 [↗](#)). These expression patterns and functional effects suggest that Kv2 conductances in many cell types might be Kv2/KvS heteromeric conductances. Consistent with narrow expression of the many KvS subunits, genetic mutations and gene-targeting studies have linked disruptions in the function of different KvS subunits to defects in distinct organ systems including retinal cone dystrophy (Wu et al., 2006 [↗](#); Hart et al., 2019 [↗](#); Inamdar et al., 2022 [↗](#)), male infertility (Regnier et al., 2017 [↗](#)), seizures (Jorge et al., 2011 [↗](#)), and changes in pain sensitivity (Tsantoulas et al., 2018 [↗](#); Lee et al., 2020b [↗](#)). These organ-specific disruptions suggest that each KvS subunit selectively modulates a subset of Kv2 channels. However, studies of the physiological roles of KvS subunits have been hindered by a lack of tools to identify native KvS conductances. Due to limited KvS pharmacology, there is little evidence that definitively ascribes native K⁺ conductances to KvS-containing channels. While studies have identified native conductances attributed to KvS subunits (reviewed by Bocksteins, 2016 [↗](#)), it has not been clear whether the Kv2 conductances that are prominent in many electrically-excitable cell types are carried by Kv2-only channels, or Kv2/KvS heteromeric channels.

No drugs are known to be selective for KvS subunits. However, Kv2/KvS heteromeric channels do have some pharmacology distinct from channels that contain only Kv2 subunits. Quaternary ammonium compounds, 4-aminopyridine, and other broad-spectrum K⁺ channel blockers have different potencies against certain KvS-containing channels as compared to Kv2 channels (Post et

al., 1996 [\[1\]](#); Thorneloe and Nelson, 2003 [\[2\]](#); Stas et al., 2015 [\[3\]](#)). However, these blockers are poorly selective and cannot effectively isolate Kv2/KvS conductances from the many other voltage-gated K⁺ conductances of electrically excitable cells.

Highly-selective Kv2 channel inhibitors fall into two mechanistically distinct classes. One class is the inhibitory cystine knot peptides from spiders. An exemplar of this class is the tarantula toxin guangxitoxin-1E (GxTX), which has remarkable specificity for Kv2 channel subunits over other voltage-gated channels (Herrington et al., 2006 [\[4\]](#); Thapa et al., 2021 [\[5\]](#)). GxTX binds to the voltage sensor of each Kv2 subunit (Milescu et al., 2009 [\[6\]](#)), and stabilizes that voltage sensor in a resting state to prevent channel opening (Tilley et al., 2019 [\[7\]](#)). GxTX binding requires a specific sequence of residues, TIFLTES, at the extracellular end of the Kv2 subunit S3 transmembrane helix (Milescu et al., 2013 [\[8\]](#)). This GxTX-binding sequence is conserved between Kv2 channels but is not retained by any KvS subunit. A second class of selective Kv2 inhibitors is a family of small molecules discovered in a high throughput screen for use-dependent Kv2 inhibitors (Herrington et al., 2011 [\[9\]](#)). Of these, RY785 is the most selective for Kv2 channels over other channel types. RY785 acts like a pore blocker which binds in the central cavity of Kv2 channels (Marquis and Sack, 2022 [\[10\]](#)). The central cavity-lining residues of all KvS subunits have differences from Kv2 subunits. We recently reported that coexpression of Kv5.1 with Kv2.1 led to a conductance that was resistant to RY785 (Ferns et al., 2025 [\[11\]](#)).

In this study, we develop a method to isolate conductances of KvS-containing channels. We identify that the combination of GxTX and RY785 can distinguish conductances of Kv2-only channels from channels that contain the KvS subtypes, Kv5, Kv6, Kv8, or Kv9. To determine whether cell types enriched with KvS mRNA have functional KvS-containing channels, we use these inhibitors to reveal native neuronal conductances consistent with Kv2/KvS heteromers in mouse and human dorsal root ganglion neurons. While this study does not address the impact of GxTX or RY785 on action potentials or *in vivo*, the distinct pharmacology of Kv2/KvS heteromers presented here suggests that KvS conductances could be targeted to selectively modulate discrete subsets of cell types.

Results

Kv2.1/Kv8.1 heteromers are resistant to RY785 and sensitive to GxTX

To identify a pharmacological strategy to distinguish Kv2/KvS heteromeric conductances from other endogenous neuronal conductances, we determined the Kv2/KvS selectivity of known Kv2 inhibitors. To test whether Kv2 inhibitors also inhibit Kv2/KvS heteromeric channels, we transfected KvS cDNA into a stable cell line which was subsequently induced to express Kv2.1 (Kv2.1-CHO) and later recorded whole cell currents. We previously found that 1 μ M RY785 or 100 nM GxTX blocked almost all the voltage-gated K⁺ conductance of this Kv2.1-CHO cell line, with $1 \pm 2\%$ or $0 \pm 0.1\%$ (mean $\pm$ SEM) current remaining respectively at 0 mV (Tilley et al., 2019 [\[7\]](#); Marquis and Sack, 2022 [\[10\]](#)). To test the pharmacological response of KvS we began with Kv8.1, a subunit that creates heteromers with biophysical properties distinct from Kv2 homomers (Salinas et al., 1997a [\[12\]](#)), and modulates motor neuron vulnerability to cell death (Huang et al., 2024 [\[13\]](#)). After transfection of Kv8.1 into Kv2.1-CHO cells, we find that a sizable component of the delayed rectifier current became resistant to 1 μ M RY785 (**Fig 1 A** [\[14\]](#)). This RY785-resistant current was inhibited by 100 nM GxTX, suggesting that the GxTX-sensitivity arises from inclusion of Kv2.1 subunits in the channels underlying the RY785-resistant current. A simple interpretation is that RY785-resistant yet GxTX-sensitive currents are carried by Kv2.1/Kv8.1 heteromeric channels. The fraction of RY785-resistant current had a pronounced cell-to-cell variability (**Fig 1 D** [\[15\]](#)). Co-

expression of KvS and Kv2 subunits can result in Kv2 homomers and Kv2/KvS heteromers (Pisupati et al., 2018), and we presume variability in the RY785-sensitive fraction results from cell-to-cell variability in the proportion of Kv8.1-containing channels.

As a control, we transfected Kv2.1-CHO cells with Navβ2, a transmembrane protein not expected to interact with Kv2.1. In Kv2.1-CHO cells transfected with Navβ2, 1 μM RY785 efficiently blocked Kv2.1 conductance, leaving $4 \pm 0.6\%$ (mean $\pm$ SEM) of current (Fig 1 B and D). We also transfected Kv2.1-CHO cells with a member of the AMIGO family of Kv2-regulating transmembrane proteins. AMIGO1 promotes voltage sensor activation of Kv2.1 channels in these Kv2.1-CHO cells (Sepela et al., 2022). 1 μM RY785 blocked Kv2.1 conductances in cells transfected with AMIGO1, leaving $0.6 \pm 1\%$ (SEM) of current (Fig 1 C and D). These control experiments indicate that transfection of a set of other transmembrane proteins did not confer resistance to RY785, suggesting that the RY785 resistance is not generically induced by overexpression of non-KvS transmembrane proteins.

To determine whether Kv8.1-containing channels are completely resistant to RY785, we performed an RY785 concentration-effect experiment. To pre-block Kv2.1 homomers we began concentration-effect measurements at 0.35 μM RY785, which we expect to block 98% of homomers based on the estimated K_D of 6 nM RY785 for the Kv2.1 currents in these Kv2.1-CHO cells (Marquis and Sack, 2022). Notably, currents resistant to 0.35 μM RY785 were blocked by higher concentrations of RY785, with nearly complete block observed in 35 μM RY785 (Fig 2 A). We quantified block of tail currents at -9 mV following a 200 ms step to +71 mV, and normalized to current from the initial 0.35 μM RY785 treatment. This protocol revealed an IC_{50} of 5 ± 1 μM (SD) (Fig 2 B). The Hill coefficient of 1.2 ± 0.2 is consistent with 1:1 binding to a homogenous population of RY785-inhibited channels. We had noted that solution exchanges can change current amplitudes, and interleaved time-matched solution exchange controls. These controls revealed variable current rundown of approximately 30% on average (Fig 2 C). Time-matched control washes were followed by treatment with 35 μM RY785 to confirm that currents in these cells had similar RY785 sensitivity to those in our concentration-effect experiment. Following block by 35 μM RY785, washing with 0.35 μM RY785 caused increases in current amplitudes in every trial (Fig 2 D). The run-down and incomplete wash-out indicates that the 5 μM IC_{50} of RY785 for these resistant channels may be an underestimate. Kv2.1/Kv8.1 currents were unblocked in the first current test following RY785 washout (Fig 2 Supplement). This rapid recovery indicates that unblocking of Kv2.1/Kv8.1 heteromers occurred during the less than 3 min wash time at -89 mV, or that RY785 unblocked on the millisecond time scale during activating voltage pulses. This is distinct from Kv2.1 homomers, where RY785 becomes trapped in deactivated channels and unblocks much more slowly, with a time constant of about 2 hours at -92 mV or 100 s at +28 mV (Marquis and Sack, 2022). The dramatically faster unblock from Kv2.1/Kv8.1 is consistent with the weaker affinity observed for RY785. Overall, the results suggest that Kv8.1-containing channels in these Kv2.1-CHO cells form a pharmacologically-homogenous population with an affinity for RY785 ~3 orders of magnitude weaker than Kv2.1 homomers. Our estimates of the affinities of Kv2.1 homomeric and Kv2.1/Kv8.1 heteromeric channels for RY785 suggest that ~1 μM RY785 elicits nearly complete block of Kv2.1 homomer conductance while blocking little Kv2.1/KvS heteromer conductance.

Biophysical properties of RY785-resistant conductance are consistent with Kv2.1/Kv8.1 channels

We wondered whether RY785 block of Kv2 homomers could better reveal the gating of heteromers. Previous studies have identified that Kv2/Kv8.1 gating kinetics are distinct from Kv2 homomers (Salinas et al., 1997a) and we set out to identify whether RY785-resistant currents from Kv8.1 transfected CHO cells had similarly modulated kinetics. While concentrations of RY785 that partially block Kv2.1/Kv8.1 modified the kinetics of voltage-dependent gating, suggesting state-dependent block, we did not observe modification of kinetics with 3.5 μM or lower

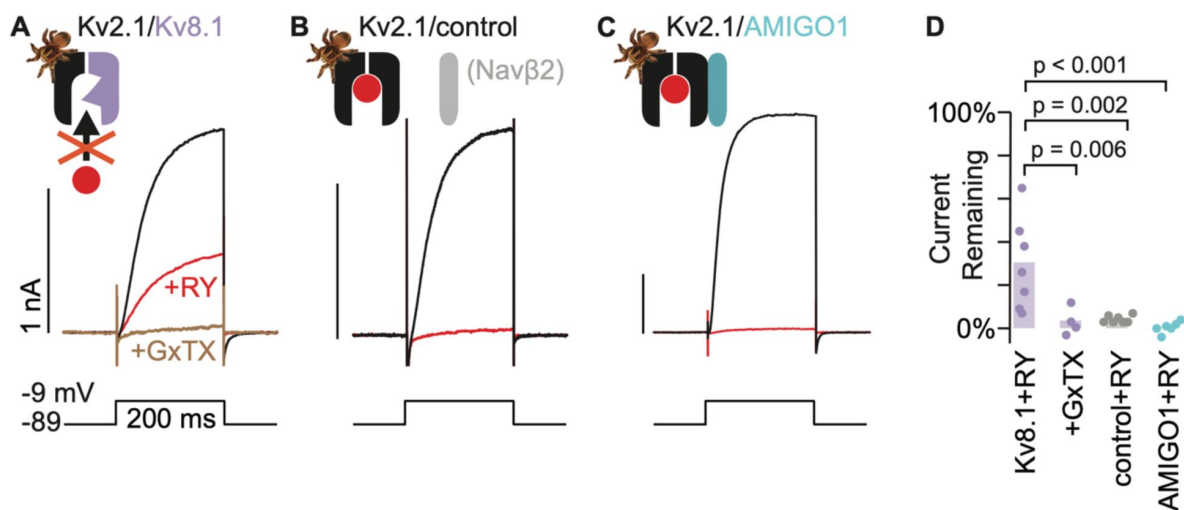


Figure 1.

Kv2.1/Kv8.1 heteromers are resistant to RY785 and sensitive to GxTX

A) Exemplar traces from a voltage-clamped Kv2.1-CHO cell transfected with Kv8.1. Black and red traces are currents before and after application of 1 μ M RY785 respectively. Brown trace is current after subsequent application of 1 μ M RY785 and 100 nM GxTX.

B) Exemplar traces from a Kv2.1-CHO cell transfected with Nav β 2.

C) Exemplar traces from a Kv2.1-CHO cell transfected with AMIGO1.

D) Current remaining after application of 1 μ M RY785 or 1 μ M RY785 + 100 nM GxTX. Bars represent mean. Each point represents current from one cell at the end of a 200 ms voltage step to -9 mV. Dunnett tests with Kv8.1 + RY785 as control.

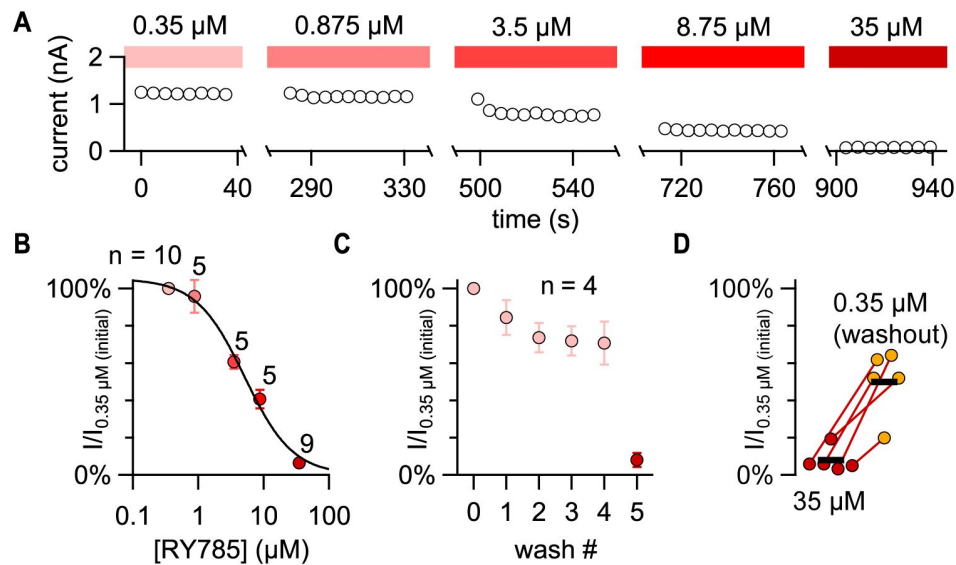


Figure 2.

RY785 blocks Kv2.1/Kv8.1 heteromers in a concentration-dependent manner.

A) Current amplitudes during an RY785 concentration-effect experiment on a Kv2.1-CHO cell transfected with Kv8.1. Circles represent tail current amplitudes 2–4 ms into a step to -9 mV following a 200 ms activating step to 71 mV. Voltage protocol was repeated in 5 s intervals. Solution exchanges occurred during the gaps in the time axis. For exemplar current traces, see [Figure 2 Supplement](#) and [Figure 3](#).

B) Mean normalized tail current amplitudes with increasing concentrations of RY785. Error bars represent SEMs. Black curve is a fitted Hill function with $n_H = 1$ ($\text{IC}_{50} = 5.1 \pm 1.0 \mu\text{M}$, base = $1.0 \pm 0.1 \%$).

C) Vehicle control tail current with repeated solution exchanges (washes) into 0.35 μM RY785, mimicking solution exchanges in panel B. Vehicle control solution exchanges were followed by exchange into 35 μM RY785 (wash #5). Error bars represent SEMs from $n = 4$ cells.

D) Tail current recovery following solution exchange from 35 μM RY785 into 0.35 μM RY785 (washout). Bars represent mean current amplitudes from $n = 5$ cells.

concentrations of RY785 (**Fig 3**). To study the biophysical properties of the Kv8.1 conductance in the Kv2.1-CHO cells, we analyzed currents in 1 μ M RY785 to block the Kv2.1 homomers. For comparison, Kv2.1-CHO cells were transfected with a control plasmid and treated with a DMSO vehicle control (**Fig 4 Supplement 1**). We stepped cells to -9 mV from a holding potential of -89 mV and fit the current rise with an exponential function (**equation 1**). Cells transfected with Kv8.1 and blocked with 1 μ M RY785 had a significantly slower activation time constant and lower sigmoidicity (shorter relative activation delay) than those only expressing Kv2.1 (**Fig 4 A-C**). Conductance-voltage relations were fit with a Boltzmann function (**equation 2**) revealing that the half-maximal conductance of Kv8.1-transfected cells is shifted positive relative to Kv2.1 alone (**Fig 4 D** and **E**). We did not detect a significant difference in the steepness (z) of the conductance-voltage relation (**Fig 4 F**). Currents from Kv8.1-transfected cells inactivated less during a 10 s step to -9 mV (**Fig 4 G** and **H**). However, the steady-state inactivation of Kv8.1-transfected cells was shifted to more negative voltages and is less steep than Kv2.1-transfected cells (**Fig 4 I-K**). Overall, the biophysical properties reported here are consistent with a previous report which identified that co-expression of Kv8.1 with Kv2.1 in *Xenopus* oocytes slows the rate of activation, reduces inactivation and shifts steady-state inactivation to more negative voltages (Salinas et al., 1997a). This previous report also identified a positive shift in the conductance-voltage relation when Kv8.1 is co-expressed with a Kv2 subunit (Kv2.2), similar to our findings with Kv2.1/Kv8.1. Together these results show that RY785-resistant currents in cells transfected with Kv8.1 are distinct from Kv2.1 homomer currents and have changes in gating consistent with prior reports of Kv8.1/Kv2 biophysics. This validates using RY785 block of Kv2 homomers as a method to reveal the biophysics of a Kv8.1-containing population.

A subunit from each KvS subtype is resistant to RY785 but sensitive to GxTX

To test if RY785 resistance and GxTX sensitivity of Kv8.1 is shared broadly by KvS subunits, we similarly assessed subunits of Kv5, Kv6, and Kv9 subtypes: Kv5.1, Kv6.4, and Kv9.3. Each of these KvS subunits create Kv2/KvS heteromers that have distinct biophysical properties (Kramer et al., 1998; Kerschensteiner and Stocker, 1999; Bocksteins et al., 2012). Kv5.1/Kv2.1 heteromers play an important role in controlling the excitability of mouse urinary bladder smooth muscle (Malysz and Petkov, 2020), mutations in Kv6.4 have been shown to influence human labor pain (Lee et al., 2020b), and deficiency of Kv9.3 disrupts parvalbumin interneuron physiology in mouse prefrontal cortex (Miyamae et al., 2021). We transfected Kv5.1, Kv6.4, and Kv9.3, to determine whether they also produced delayed rectifier current resistant to 1 μ M RY785 yet sensitive to 100 nM GxTX (**Fig 5 A-C**). We observed that, in 1 μ M RY785, >10% of the voltage-gated current remained in 12/13 Kv5.1, 9/14 Kv6.4, and 5/5 Kv9.3 transfected cells (**Fig 5 D**). Like Kv8.1, the fraction of RY785-resistant current had pronounced cell-to-cell variability (**Fig 5 D**) suggesting that the RY785-sensitive fraction could be due to different ratios of functional Kv2.1 homomers to Kv2.1/KvS heteromers. Addition of 100 nM GxTX blocked RY785-resistant current from cells transfected with each of these KvS subunits. A slightly higher fraction of Kv9.3 current remained in 100 nM GxTX, possibly due to Kv9.3 negatively shifting the midpoint of the conductance voltage relationship (Kerschensteiner and Stocker, 1999). While a fraction of KvS subunits appear to be retained intracellularly, immunofluorescence for Kv5.1, Kv9.3 and Kv2.1 also appeared localized to the perimeter of transfected Kv2.1-CHO cells (**Figure 1 Supplement**). These results show that voltage-gated outward currents in cells transfected with members from each KvS subtype have decreased sensitivity to RY785 but remain sensitive to GxTX. While we did not test every KvS subunit, the ubiquitous resistance suggests that all KvS subunits may provide resistance to 1 μ M RY785 yet remain sensitive to GxTX, and that RY785 resistance is a hallmark of KvS-containing channels.

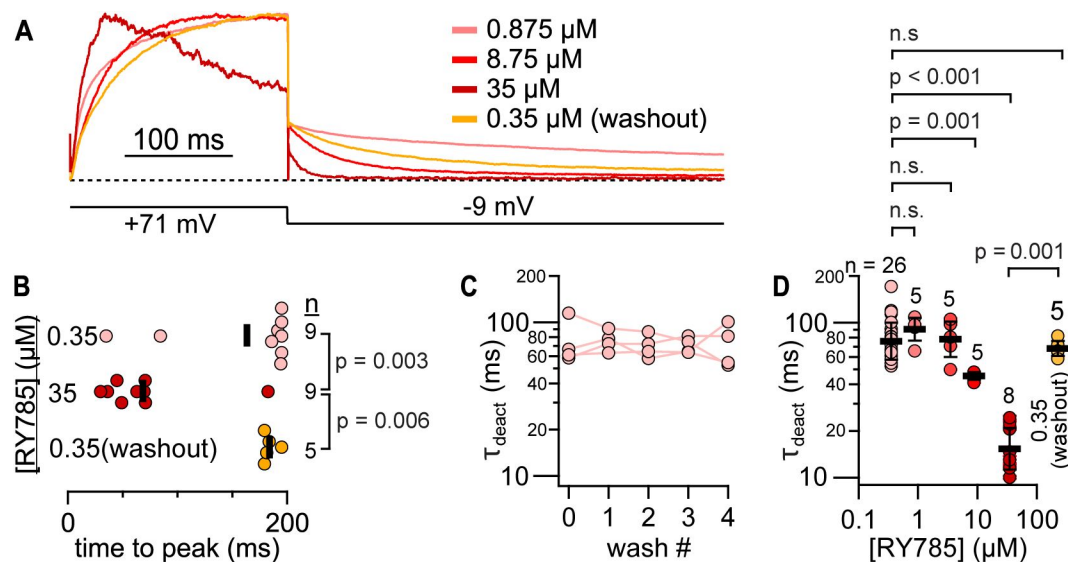


Figure 3.

RY785 can affect Kv2.1/Kv8.1 current kinetics. 0.1 1 10 100 [RY785] (μM)

A) Kinetics of currents from a Kv2.1-CHO cell transfected with Kv8.1 are altered by RY785. Traces normalized to max.

B) Latency to peak current during steps to +71 mV. The time axis of this plot is aligned with that of Panel A. Bars represent means. Unpaired Wilcoxon rank tests.

C) Time constant of deactivation at -9 mV is constant after washes with 0.35 μM RY785. Time constants are derived from fits of a monoexponential function (Equation 3) with A_2 set equal to 0 to tail currents like those shown in Panel A. Fits were from the peak of each tail current to 200 ms after the voltage step. Brown and Forsythe test $p = 0.98$. ANOVA $p = 0.98$. Statistics were performed on natural logarithms of time constants. $n = 4$ cells.

D) RY785 can alter time constant of deactivation. Bars represent means. Brown and Forsythe test $p = 3 \times 10^{-12}$. Unpaired Welch test $p = 1 \times 10^{-7}$. Dunnett tests with 0.35 μM RY785 (initial) as control. Unpaired Wilcoxon rank test comparing 35 μM RY785 to 0.35 μM RY785 (washout) $p = 0.001$. Statistics were performed on natural logarithms of time constants.

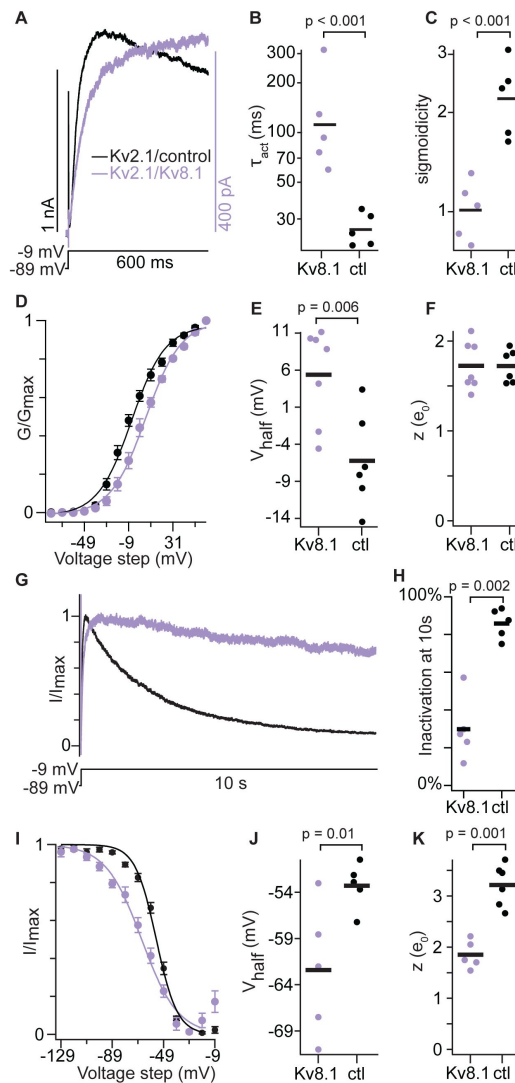


Figure 4.

RY785-resistant current is consistent with Kv2.1/Kv8.1 heteromers

Kv2.1/8.1 data (purple) are from Kv2.1-CHO cells transfected with Kv8.1, and are in 1 μ M RY785. Kv2.1/control (black) were transfected with Nav β 2, and are in vehicle control solution. Before measurements, repeated voltage steps to -9 mV were given until currents stabilized. p values are from two-tailed unpaired Wilcoxon rank test.

A) Exemplar currents during a step to -9 mV.

B) Time constants from exponential fit (Equation 1). Bars represent geometric mean.

C) Sigmoidicity from exponential fit (Equation 1). Bars represent geometric mean.

D) Conductance-voltage activation relation. Conductance was measured from initial tail currents at -9 mV. Mean $\pm$ SEM.

Kv2.1/Kv8.1 $n = 7$ cells Kv2.1 $n = 6$ cells. Lines are Boltzmann fits (Equation 2) (Kv2.1/Kv8.1: $V_{1/2} = 6 \pm 1$ mV, $z = 1.6 \pm 0.1 e_0$; Kv2.1/control: $V_{1/2} = -6.3 \pm 1$ mV, $z = 1.7 \pm 0.1 e_0$).

E) Activation $V_{1/2}$ values from individual cells. Bars represent means.

F) Activation z values. Bars represent means.

G) Exemplar currents during a 10 s step to -9 mV.

H) Percent of current inactivated after 10 seconds at -9 mV. Bars represent means.

I) Steady state currents at -9 mV after holding at indicated voltages for 10 seconds. Normalized to the max and min. Mean $\pm$ SEM. Kv2.1/Kv8.1 $n = 5$ cells Kv2.1 $n = 5$ cells. Lines are Boltzmann fits (Equation 2) (Kv2.1/Kv8.1: $V_{1/2} = -66 \pm 1$ mV, $z = 1.8 \pm 0.1 e_0$; Kv2.1/control: $V_{1/2} = -54.7 \pm 0.8$ mV, $z = 3.1 \pm 0.3 e_0$).

J) Inactivation $V_{1/2}$ values from individual cells. Bars represent means.

K) Inactivation z values. Bars represent means.

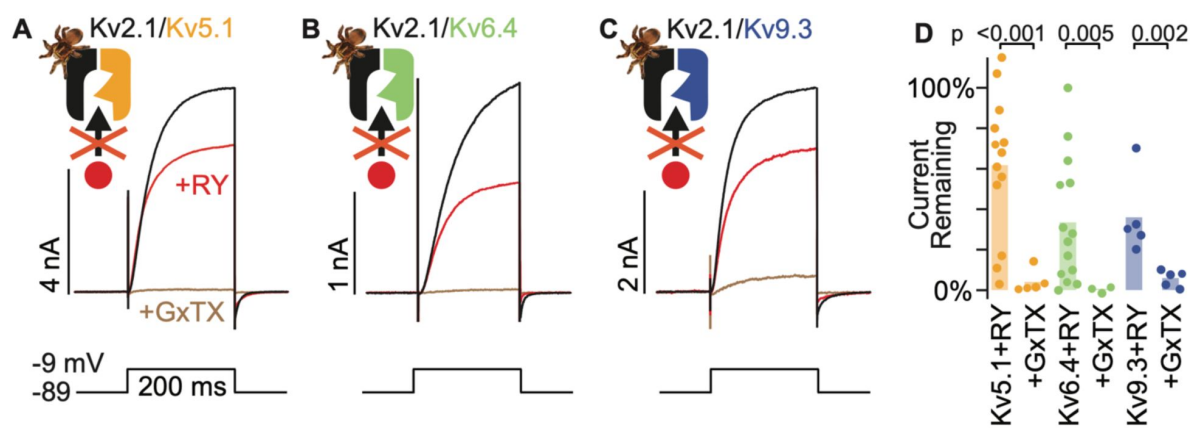


Figure 5.

A subunit from each KvS subtype is resistant to RY785

A) Exemplar traces from a voltage-clamped Kv2.1-CHO cell transfected with Kv5.1. Black and red traces are currents before and after application of 1 μ M RY785 respectively. Brown trace is current after subsequent application of 1 μ M RY785 + 100 nM GxTX

B) Exemplar traces from a Kv2.1-CHO cell transfected with Kv6.4.

C) Exemplar traces from a Kv2.1-CHO cell transfected with Kv9.3.

D) Current remaining after application of 1 μ M RY785 or 1 μ M RY785 + 100 nM GxTX. Bars represent mean. Each point represents current from one cell at the end of a 200 ms voltage step to -9 mV. Unpaired Wilcoxon rank tests.

The Kv2 conductances of mouse superior cervical ganglion neurons do not have KvS-like pharmacology

We set out to assess whether RY785 together with GxTX could be a means of distinguishing endogenous Kv2/KvS heteromers from Kv2 channels in native neurons. We first designed experiments to test whether RY785 could inhibit endogenous Kv2 currents in mice, by studying neurons unlikely to express KvS subunits. Rat superior cervical ganglion (SCG) neurons have robust GxTX-sensitive conductances (Liu and Bean, 2014) yet transcriptomics have revealed little evidence of KvS expression (Sapio et al., 2020). We investigated whether SCG neurons have conductances consistent with Kv2/KvS heteromers. As functional characterization alone cannot be trusted to classify their channel mediators of conductances, we define conductances consistent with Kv2/KvS heteromers as 'KvS-like' and conductances consistent with Kv2 homomers as 'Kv2-like'. To help isolate Kv2-like and KvS-like currents, we bathed SCG neurons in a cocktail of Nav, Cav, Kv1, Kv3, and Kv4 inhibitors then recorded voltage-gated currents. We found that exposing SCG neurons to 1 μ M RY785 inhibited most of the voltage-gated current, and subsequent addition of 100 nM GxTX to the same neurons inhibited little additional current (Fig 6 A and B). To quantify inhibition, we analyzed tail currents 10 ms after repolarizing to -45 mV. This time window was chosen because a hallmark of Kv2 currents is relatively slow deactivation (Thorneloe and Nelson, 2003; Zheng et al., 2019). Tail currents after application of 1 μ M RY785 were decreased by 88 ± 5 % (mean $\pm$ SEM) in SCG neurons (Fig 6 B). Subsequent application of 100 nM GxTX had little further effect. To determine if the RY785-sensitive conductances are consistent with previous reports of Kv2 channels, we examined the biophysical properties of the Kv2-like (RY785-sensitive) currents defined by subtraction (Fig 6 C). Current activation began to be apparent at -45 mV and had a conductance that was half maximal at -11 mV (Fig 6 D). The faster component of deactivation of Kv2-like currents in SCG neurons had a time constant of $16 \text{ ms} \pm 0.6$ (mean $\pm$ SEM) at -45 mV (Fig 6 E). These results are consistent with reported biophysical properties of Kv2 channels (Kramer et al., 1998; Liu and Bean, 2014; Tilley et al., 2019; Sepela et al., 2022). Together these results show that 1 μ M RY785 almost completely inhibits endogenous Kv2-like conductances in these mouse neurons, consistent with mouse SCG neurons having few, if any, functional Kv2/KvS heteromers. We cannot rule out that the small amount of current remaining after RY785 (12% of the control) is due to Kv2/KvS heteromers, but its insensitivity to GxTX suggests that it may instead be current from a non-Kv2 channel remaining in the cocktail of K-channel inhibitors.

The Kv2 conductances of mouse dorsal root ganglion neurons have KvS-like pharmacology

To determine if RY785/GxTX pharmacology could reveal endogenous KvS-containing channels, we next studied neurons likely to express KvS subunits. Mouse dorsal root ganglion (DRG) somatosensory neurons express Kv2 proteins (Stewart et al., 2024), have GxTX-sensitive conductances (Zheng et al., 2019), and express a variety of KvS transcripts (Bocksteins et al., 2009; Zheng et al., 2019), yet transcript abundance does not necessarily correlate with functional protein abundance. To record from a consistent subpopulation of mouse somatosensory neurons which has been shown to contain GxTX-sensitive currents and have abundant expression of KvS mRNA transcripts (Zheng et al., 2019), we used a *Mrgprd*^{GFP} transgenic mouse line which expresses GFP in nonpeptidergic nociceptors (Zylka et al., 2005; Zheng et al., 2019). Deep sequencing identified that mRNA transcripts for Kv5.1, Kv6.2, Kv6.3, and Kv9.1 are present in GFP⁺ neurons of this mouse line (Zheng et al., 2019) and we confirmed the presence of Kv5.1 and Kv9.1 transcripts in GFP⁺ neurons from *Mrgprd*^{GFP} mice using RNAscope (Fig 7 Supplement 1). We investigated whether these neurons have conductances consistent with KvS-like pharmacology by performing whole cell voltage clamp on cultured DRG neurons that had clear GFP fluorescence. Voltage-clamped neurons were bathed in the same

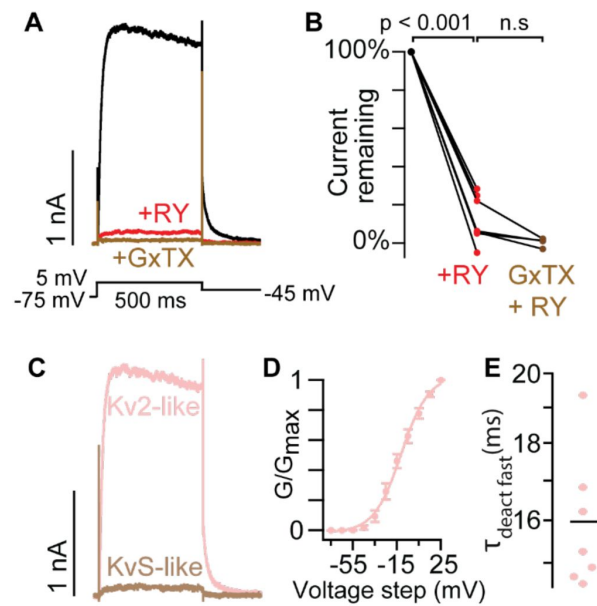


Figure 6.

The Kv2 conductances of mouse superior cervical ganglion neurons do not have KvS-like pharmacology.

- A) Exemplar currents from a voltage-clamped SCG neuron. Black and red traces are currents before and after application of 1 μ M RY785 respectively. Brown trace is current after subsequent application of 1 μ M RY785 + 100 nM GxTX.
- B) Tail current amplitude 10 ms after voltage was stepped from +5 mV to -45 mV normalized to current amplitude before RY785. Paired Wilcoxon rank tests, $n = 7$ neurons, $N = 3$ mice.
- C) Subtracted currents from A. Kv2-like current is the RY785-sensitive current (black trace minus red in A). KvS-like current is the GxTX-sensitive current remaining in RY785 (red trace minus brown in A).
- D) Conductance-voltage activation relation of Kv2-like current in SCG neurons. Conductance was measured from tail currents at -45 mV. $V_{1/2} = -11 \pm 1$ mV, $z = 2.1 \pm 0.2 e_0$ Mean $\pm$ SEM. $n = 7$ neurons, $N = 3$ mice.
- E) The faster time constant of a double exponential (Equation 3) fit to channel deactivation at -45 mV. Bar represents mean.

cocktail of channel inhibitors used on SCG neurons, plus the Nav1.8 inhibitor A-803467. Application of 1 μ M RY785 inhibited outward currents somewhat, but unlike in SCG neurons, a prominent delayed-rectifier outward conductance with slow deactivation remained (**Fig 7 A** [left panel](#)). Tail currents in 1 μ M RY785 decreased $29 \pm 3\%$ (mean $\pm$ SEM) (**Fig 7 B** [left panel](#)). Subsequent application of 100 nM GxTX decreased tail currents by $68 \pm 5\%$ (mean $\pm$ SEM) of their original amplitude before RY785. We do not know the identity of the outward current that remains in the cocktail of inhibitors + RY785 + GxTX. We observed variable current run-up or run-down but no significant effect of vehicle in blinded, interleaved experiments, while RY785 significantly decreased tail currents relative to vehicle controls (**Fig 7 A** [and B](#) [right panel](#)). We do not know what conductances the vehicle solution exchange affects, the changes appear to be time-dependent or due to the solution exchange itself. Concurrent application of 100 nM GxTX and 1 μ M RY785 to neurons in vehicle decreased currents by $69 \pm 5\%$ (mean $\pm$ SEM).

To determine if the RY785- and GxTX-sensitive conductances in GFP⁺ neurons from *Mrgprd*^{GFP} mice are consistent with previous reports of Kv2 homomeric or Kv2/KvS heteromeric channels, we examined the biophysical properties of the Kv2-like (RY785-sensitive) and KvS-like (RY785-resistant, GxTX-sensitive) currents defined by subtraction (**Fig 7 C** [left panel](#)). Obvious Kv2-like and KvS-like channel conductances began at -44 mV and had half-maximal conductances around -19 mV (**Fig 7 D** [left panel](#)), consistent with Kv2 and many KvS-containing channels (Kramer et al., 1998 [left panel](#); Richardson and Kaczmarek, 2000 [left panel](#); Sano et al., 2002 [left panel](#); Thorne and Nelson, 2003 [left panel](#)). While changes in inactivation are prominent with KvS subunits, we did not investigate inactivation in neurons because the lengthy depolarizations required often resulted in irreversible leak current increases that degraded the accuracy of RY785/GxTX subtraction current quantification. We did note that the KvS-like currents deactivated slower than Kv2-like currents (**Fig 7 E** [left panel](#)), consistent with the effects of several KvS subunits whose transcripts are expressed in nociceptor DRG neurons. Kv5.1, Kv6.3, and Kv9.1 all slow deactivation of Kv2 conductances in heterologous cells (Salinas et al., 1997b [left panel](#); Kramer et al., 1998 [left panel](#); Sano et al., 2002 [left panel](#)), and in Kv2.1-CHO cells transfected with Kv5.1, we confirmed that RY785-resistant currents deactivate slower than Kv2.1 controls (**Fig 7 Supplement 2** [left panel](#)). Together, these results indicate that, in these mouse DRG neurons, RY785-sensitive currents are Kv2-like, while RY785-resistant yet GxTX-sensitive currents are KvS-like. Under these conditions, $58 \pm 3\%$ (mean $\pm$ SEM) of the delayed rectifier conductance was resistant to RY785 yet sensitive to GxTX (KvS-like) (**Fig 7 F** [left panel](#)). We note that the ratio of KvS-to Kv2-like conductances is expected to vary with holding potential, as KvS subunits can change the degree and voltage-dependence of steady state inactivation (e.g. **Fig 4I** [left panel](#)).

We also tested the other major mouse C-fiber nociceptor population, peptidergic nociceptors, to determine if this subpopulation also has conductances resistant to RY785 yet sensitive to GxTX. We voltage clamped DRG neurons from a *CGRP*^{GFP} mouse line that expresses GFP in peptidergic nociceptors (Gong et al., 2003 [left panel](#)). Deep sequencing has identified mRNA transcripts for Kv6.2, Kv6.3, Kv8.1 and Kv9.3 present in GFP⁺ neurons, an overlapping but distinct set of KvS subunits from the *Mrgprd*^{GFP} non-peptidergic population (Zheng et al., 2019 [left panel](#)). In GFP⁺ neurons from *CGRP*^{GFP} mice, we found that a fraction of outward current was inhibited by 1 μ M RY785 and additional current inhibited by 100 nM GxTX (**Fig 7 Supplement 3 A-C** [left panel](#)). In these experiments, $58 \pm 2\%$ (mean $\pm$ SEM) was KvS-like (**Fig 7 Supplement 3 D** [left panel](#)) identifying that KvS-like conductances are present in these peptidergic nociceptors. For *CGRP*^{GFP} neurons we did not include the Kv1, Kv3, Kv4, Nav and Cav channel inhibitor cocktail used for other neuron experiments, indicating that the cocktail of inhibitors is not required to identify KvS-like conductances. Overall, these results show that the unique pharmacology of RY785 and GxTX can reveal endogenous KvS-like conductances in the major subtypes of mouse C-fiber nociceptors.

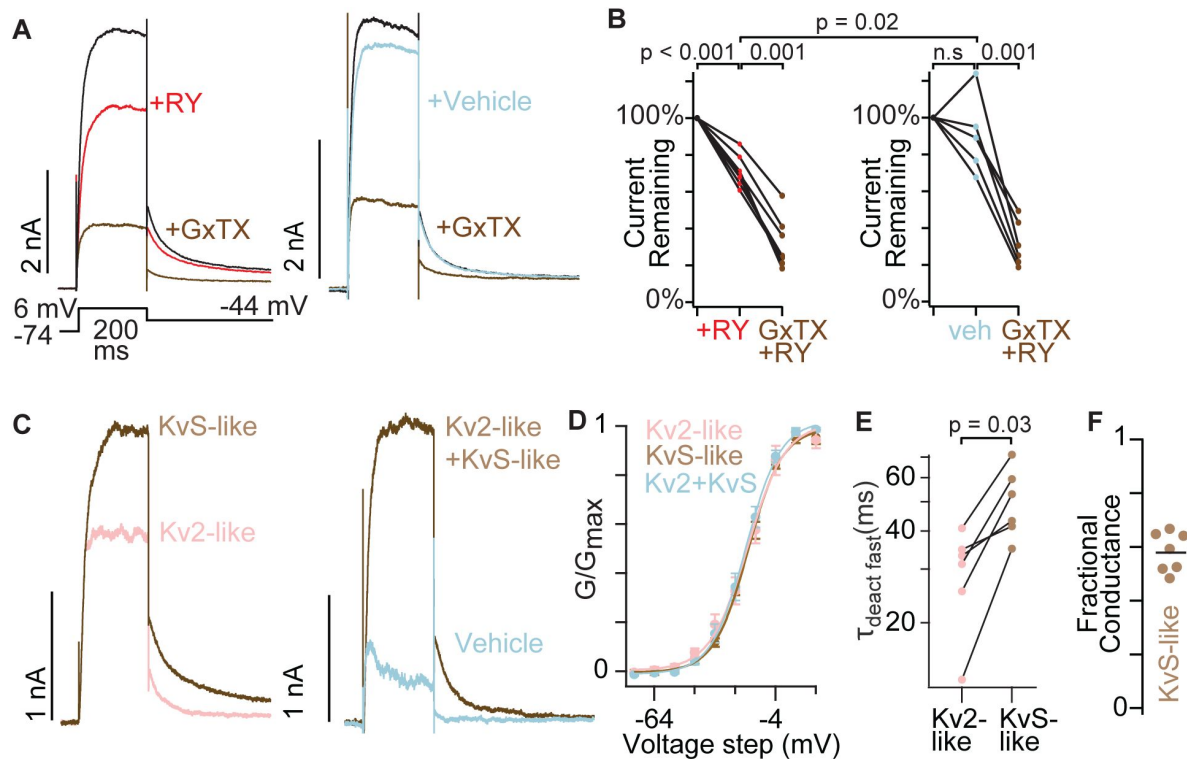


Figure 7.

The Kv2 conductances of mouse dorsal root ganglion neurons have KvS-like pharmacology.

A) Exemplar currents from nonpeptidergic nociceptors, GFP⁺ neurons from *Mrgprd*^{GFP} mice.

B) Tail current amplitude 10 ms after voltage was stepped from +6 mV to -44 mV normalized to current amplitude before RY785 or vehicle treatment. Wilcoxon rank tests were paired, except the comparison of RY785 to vehicle which was unpaired. RY785 then GxTX: n = 7 neurons, N = 4 mice. Vehicle then GxTX: n = 6 neurons, N = 4 mice.

C) Exemplar subtracted currents from A. Kv2-like is the initial current minus RY785 (black trace minus red in A left panel). KvS-like is the current in RY785 minus GxTX (red trace minus brown in A left panel). Kv2-like+KvS-like is the current in vehicle minus RY785 + GxTX (blue trace minus brown in A right panel).

D) Voltage dependence of activation of subtraction currents in *Mrgprd*^{GFP} neurons. Pink points represent Kv2-like currents, brown points represent KvS-like currents, and light blue points represent Kv2+KvS-like currents after vehicle treatment. Conductance was measured from initial tail currents at -44 mV. Kv2-like: $V_{1/2} = -18 \pm 1$ mV, $z = 2.7 \pm 0.3$ e₀, KvS-like: $V_{1/2} = -18 \pm 1$ mV, $z = 3 \pm 0.2$ e₀, Kv2 + KvS-like: $V_{1/2} = -19 \pm 1$ mV, $z = 2.9 \pm 0.1$ e₀. Mean $\pm$ SEM. KvS-like and Kv2-like n = 7 neurons N = 4 mice, Kv2+KvS-like n = 6 neurons N = 4 mice.

E) The faster time constant of a double exponential fit (Equation 3) to channel deactivation at -44 mV. p value represents paired Wilcoxon rank test.

F) Fractional KvS-like conductance relative to the total RY785 + GxTX-sensitive conductance. KvS-like is only sensitive to GxTX. Bar represents mean.

The Kv2 conductances of human dorsal ganglion neurons have KvS-like pharmacology

Human DRG neurons express Kv2 proteins (Stewart et al., 2024 [↗](#)), and express KvS transcripts (Ray et al., 2018 [↗](#)) suggesting that they may have Kv2/KvS conductances. We performed whole-cell voltage clamp on cultured human DRG neurons, choosing smaller-diameter neurons and using the same solutions as *Mrgprd*^{GFP} mouse DRG neuron recordings. In human DRG neurons, a fraction of outward current was inhibited by 1 μ M RY785 and additional current was inhibited by 100 nM GxTX, consistent with the presence of KvS-like conductances (Fig 8 A, B, C [↗](#)). These RY785- or GxTX-sensitive conductances became apparent near -44 mV and were half-maximal between -14 and -4 mV (Fig 8 D [↗](#)). Of the total conductance sensitive to RY785 + GxTX in these human DRG neurons, 76 ± 2 % (mean $\pm$ SEM) was KvS-like (Fig 8 E [↗](#)). Unlike mouse neurons, we did not detect a significant difference in tail currents of RY785 versus vehicle controls. However, RY785-subtracted currents always had Kv2-like biophysical properties whereas vehicle-subtraction currents had variable properties that precluded the same biophysical analysis. Overall, these results show that human DRG neurons can produce endogenous voltage-gated currents with pharmacology and gating consistent with Kv2/KvS heteromeric channels.

Discussion

These results identify a method for pharmacologically isolating conductances of Kv2/KvS heteromers and Kv2-only channels. RY785 blocks homomeric Kv2 channels, and subsequent application of GxTX selectively inhibits Kv2/KvS heteromeric channels. Such a protocol can aid in identification of Kv2/KvS conductances separately from the Kv2 homomer conductances that are likely to be in the same cell. Characterization of these now separable conductances can reveal impacts of Kv2 channels and KvS-containing channels on electrophysiological signaling. This is valuable as there are few other tools to probe the contributions of KvS subunits to electrical signaling in native cells and tissues.

It is remarkable that resistance to RY785 is shared across all of the KvS subtypes. We found Kv2.1/Kv8.1 conductances to be $\sim$ 1000 times less sensitive to RY785 than Kv2.1 homomer conductances in the same cell line. Based on the observation that $>50\%$ of Kv current was resistant to 1 μ M RY785 in some Kv2.1-CHO cells transfected with Kv5.1, Kv6.4, or Kv9.3, these Kv2.1/KvS channels are expected to have $IC_{50} > 1$ μ M, at least 100-fold less sensitive than the $\sim$ 6 nM IC_{50} for Kv2.1 in this cell line. This suggests that the RY785 inhibitory site is substantially disrupted by KvS subunits. We analyzed computational structural models of RY785 docked to a Kv2.1 homomer and a 3:1 Kv2.1:Kv8.1 heteromer (Fig 9 [↗](#)) to gain structural insight into how KvS subunits might interfere with RY785 binding. We used Rosetta to dock RY785 to a cryo-EM structure of a Kv2.1 homomer in an apparently open state (Fernández-Mariño et al., 2023). The top-scoring docking pose has RY785 positioned below the selectivity filter and off-axis of the pore (Fig 9 A [↗](#)), similar to a stable pose observed in molecular dynamic simulations (Zhang et al., 2024). In this pose, RY785 contacts a collection of Kv2.1 residues that vary in every KvS subtype (Fig 9 B,D,E [↗](#)). Notably, RY785 bound similarly to a 3:1 model of Kv2.1/Kv8.1, in contact with the three Kv2.1 subunits, yet avoided the Kv8.1 subunit (Fig 9 C [↗](#)). This is consistent with RY785 binding less well to Kv2.1/Kv8.1 heteromers, and also suggests that a 3:1 Kv2:KvS channel could retain a RY785 binding site when open. However, the RY785 resistance of Kv2/KvS heteromers may primarily arise from perturbed interactions with the constricted central cavity of closed channels. In homomeric Kv2.1, RY785 becomes trapped in closed channels and prevents their voltage sensors from fully activating, indicating that RY785 must interact differently with closed channels (Marquis and Sack, 2022 [↗](#)). Here we found that Kv2.1/Kv8.1 current rapidly recovers following washout of RY785, suggesting that Kv2.1/Kv8.1 heteromers do not readily trap RY785 (Figure 2

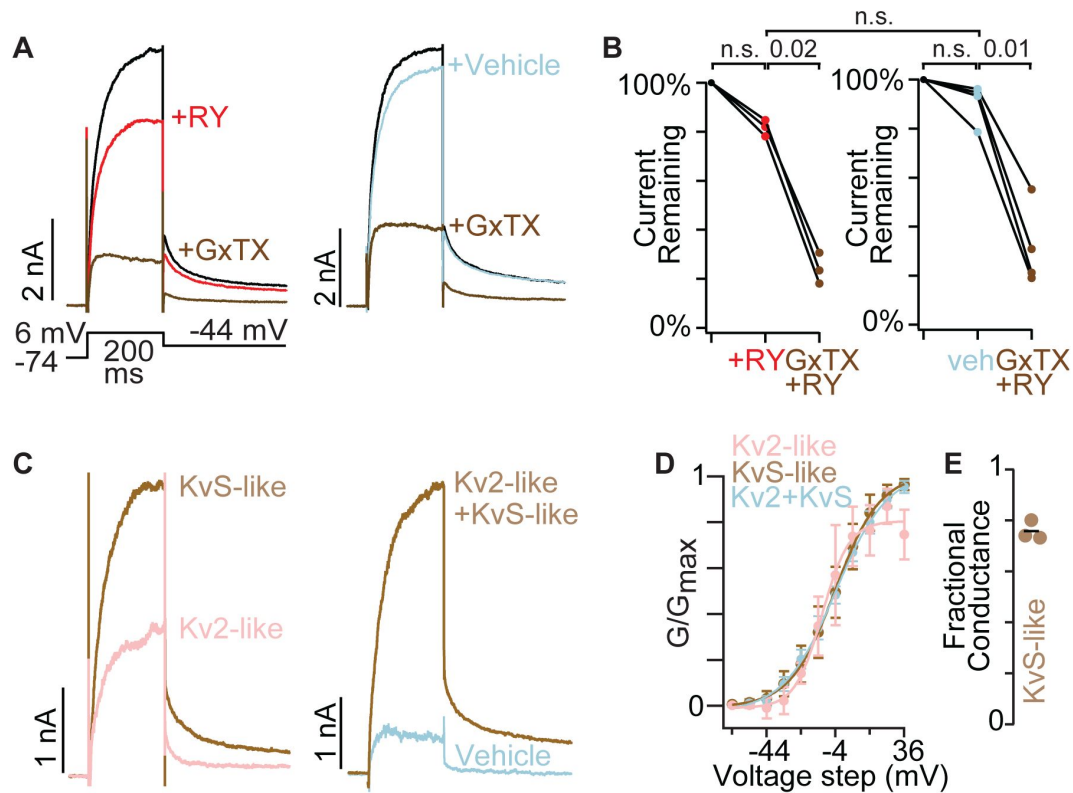


Figure 8.

The Kv2 conductances of human dorsal root ganglion neurons have KvS-like pharmacology.

- A) Exemplar currents from human dorsal root ganglion neurons.
- B) Tail current amplitude 10 ms after voltage was stepped from +6 mV to -44 mV normalized to current amplitude before RY785 or vehicle treatment. Wilcoxon rank tests were paired. RY785 then GxTX: $n = 3$ neurons. Vehicle then RY785 and GxTX: $n = 4$ neurons. All neurons from same human.
- C) Exemplar subtracted currents from A. Kv2-like is the initial current minus RY785 (black trace minus red in A left panel). KvS-like is the current in RY785 minus GxTX (red trace minus brown in A left panel). Kv2+KvS-like is the current in vehicle minus RY785 + GxTX (blue trace minus brown in A right panel).
- D) Voltage dependence of activation of subtraction currents in human dorsal root ganglion neurons. Pink points represent Kv2-like currents, brown points represent KvS-like currents, and blue points represent Kv2 + KvS-like currents after vehicle treatment. Conductance was measured from initial tail currents at -44 mV. Mean $\pm$ SEM. Kv2 + KvS-like $n = 3$ neurons $N = 1$ human, KvS+Kv2-like $n = 4$ neurons $N = 1$ human.
- E) Fractional KvS-like conductance relative to the total RY785 + GxTX-sensitive conductance. KvS-like is only sensitive to GxTX. Bar represents mean.

Supplement [↗](#)). Overall, the structural modeling suggests that KvS subunits sterically interfere with RY785 binding to the central cavity, while functional data suggest KvS subunits disrupt RY785 trapping in closed states.

While we have identified a potential means to isolate conductances from Kv2 channels containing KvS subunits, it is important to consider the limitations of these findings:

First, although every KvS subunit we tested makes conductance resistant to RY785, including Kv5.1 the most similar to Kv2.1 ([Jegla et al., 2024](#) [↗](#); [Simonson et al., 2025](#) [↗](#)) and spanning all the KvS subtypes (Kv5, Kv6, Kv8, and Kv9), we have not tested all KvS subunits, species variants, cell types, or voltage regimens. Any of these could alter the RY785 IC₅₀. Pharmacology can yield surprises, such as the unexpected resistance of human Nav1.7 to saxitoxin ([Walker et al., 2012](#) [↗](#)). The degree of resistance to RY785 may vary among KvS subunits as with other central cavity drugs such as tetraethylammonium and 4-aminopyridine ([Post et al., 1996](#) [↗](#); [Thorneloe and Nelson, 2003](#) [↗](#); [Stas et al., 2015](#) [↗](#)). Also, the degree of resistance to RY785 may vary if Kv2:KvS subunit stoichiometry varies. With high doses of RY785, we found that the concentration-response characteristics of Kv2.1/Kv8.1 in CHO cells revealed hallmarks of a homogenous channel population with a Hill slope close to 1 (**Fig 2B** [↗](#)). However, other KvS subunits might assemble in multiple stoichiometries and result in pharmacologically-distinct heteromer populations.

Second, it is possible that RY785 can modulate other voltage-gated channels. However, we think RY785 is unlikely to have substantial off-target effects, as RY785 is much less potent against other non-Kv2 voltage-gated potassium channel subfamilies as well as voltage-gated sodium and calcium channels ([Herrington et al., 2011](#) [↗](#)).

Third, concentrations of RY785 which partially blocked Kv2.1/Kv8.1 modified the gating of the voltage-activated conductance, indicating that RY785 can alter properties of heteromer conductance. Also, channel state-dependent binding is expected to influence the affinity of RY785. These effects could create complications in analyzing current subtractions.

Fourth, these are functional characterizations of currents and should not be solely trusted to classify the molecular identity of the underlying channels. While it seems unlikely that pore-forming subunits other than KvS subunits complex with Kv2 subunits, other factors could potentially disrupt the RY785 pharmacology as KvS subunits do. Regulation of Kv2 channel gating could potentially allosterically disrupt RY785 inhibition. Extensive homeostatic regulation of Kv2.1 gating maintains neuronal excitability ([Misonou et al., 2006](#) [↗](#)); for example, ischemia ([Misonou et al., 2005](#) [↗](#); [Aras et al., 2009](#) [↗](#)), glutamate ([Misonou et al., 2008](#) [↗](#)), phosphorylation ([Murakoshi et al., 1997](#) [↗](#)), SUMOylation ([Plant et al., 2011](#) [↗](#)) and AMIGO auxiliary subunits ([Peltola et al., 2011](#) [↗](#); [Maverick et al., 2021](#) [↗](#)) all alter Kv2 gating. Intriguingly, some KvS subunits are reported to functionally interact with Kv7 subunits ([Renigunta et al., 2024](#) [↗](#)). Also, the cocktail of inhibitors used in most neuron experiments here could potentially alter RY785 or GxTX action against KvS/Kv2 channels.

Despite these possibilities, we think the most parsimonious interpretation of RY785-resistant, GxTX-sensitive conductances is that they are produced by heteromeric Kv2/KvS channels. However, it is important to consider these other possibilities when interpreting RY785 resistance of GxTX-sensitive conductances. With these caveats in mind, we suggest that RY785-resistance combined with GxTX-sensitivity is strong evidence for heteromeric Kv2/KvS currents.

Delayed-rectifier Kv2-like conductances are prominent in many electrically excitable cell types. While we found only RY785-sensitive Kv2-like conductances in SCG neurons, Kv2/KvS heteromer-like conductances were dominant in DRG neurons. This striking contrast is consistent with KvS transcript abundances ([Zheng et al., 2019](#) [↗](#); [Sapio et al., 2020](#) [↗](#)). Beyond these cell types, it is unclear how prevalent Kv2/KvS heteromer conductances are. While Kv2 subunits are broadly

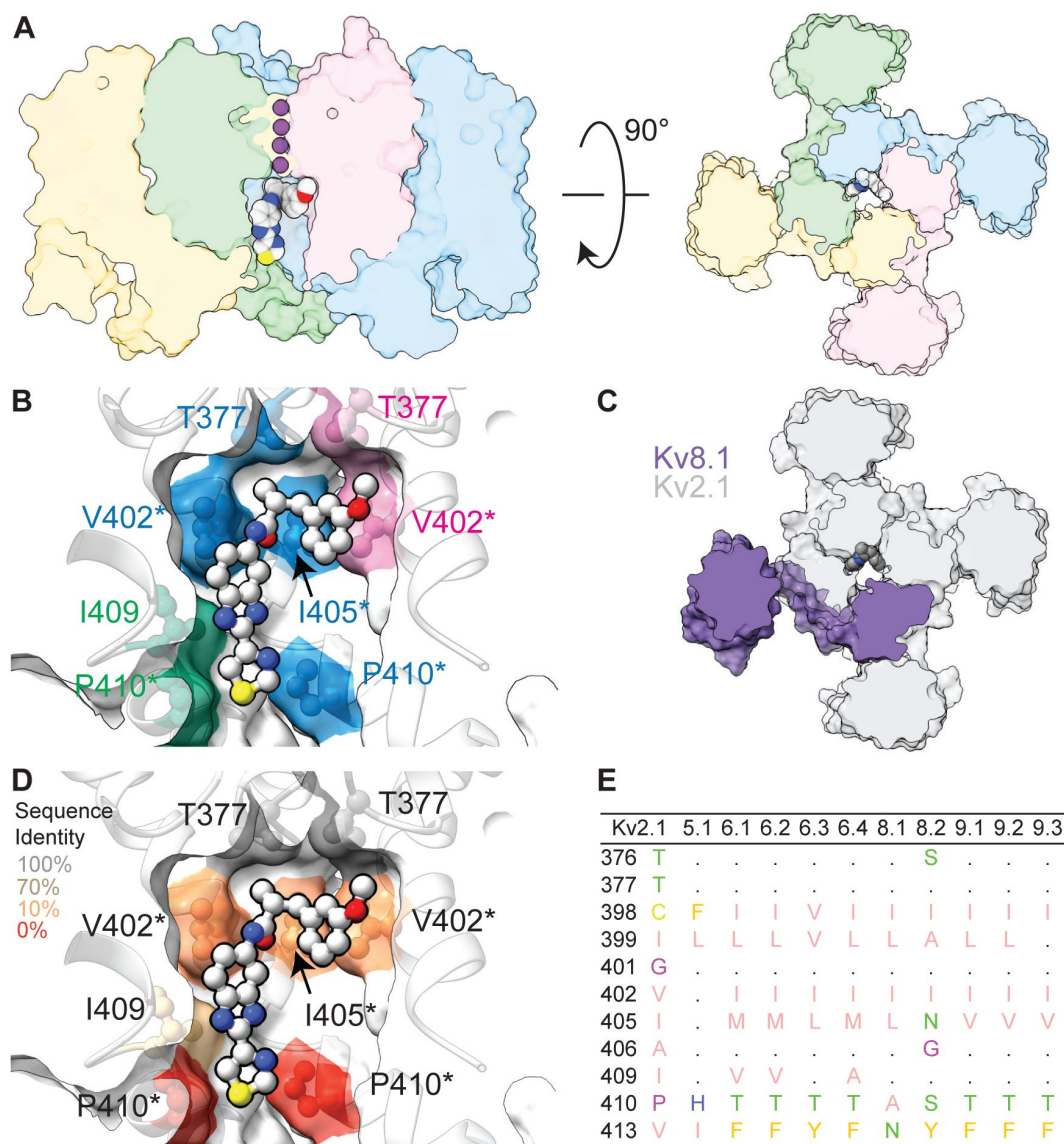


Figure 9.

Structural models of RY785 docked to a Kv2.1 homomer and 3:1 Kv2.1:Kv8.1 heteromer.

A) Model of RY785 docked to a Kv2.1 homomer (PDB 8SD3). Left: transmembrane view. Right: Extracellular view. Individual subunits colored distinctly. RY785 represented by the Corey–Pauling–Koltun space-filling model and coloring scheme. Potassium (purple spheres) from PDB 8SD3 are shown in selectivity filter.

B) A zoom in of central cavity in transmembrane view. Colors indicate subunit of residues making Van der Waals contacts with RY785. Human Kv2.1 sequence numbering. Asterisks indicate residues that vary in Kv8.1.

C) Model of RY785 docked to a 3:1 Kv2.1:Kv8.1 heteromer. RY785 makes no contact with Kv8.1.

D) A replica of panel B with heatmap coloring to indicate sequence conservation of Kv2.1 to KvS subunits.

E) Sequence comparison of pore-lining S6 residues of Kv2.1 and all human KvS subunits. Zappo coloring indicates physiochemical identity. A dot indicates sequence conservation with Kv2.1.

expressed in electrically excitable cells throughout the brain and body, transcripts for KvS subunits have unique expression patterns that are specific to each KvS subtype (Bocksteins, 2016) and can fluctuate with age (Regnier et al., 2016). However, transcript levels alone are not sufficient to predict protein levels (Liu et al., 2016). We found recently that native Kv2 channels in mouse brain contain KvS subunit proteins, including Kv5.1, Kv8.1, Kv9.1 and Kv9.2. Notably, the KvS mass spectral abundance relative to Kv2.1 ranged from $\approx 18\%$ for Kv5.1 to 2% for Kv9.1 (Ferns et al., 2025), and Kv5.1 protein expression was largely restricted to cortical neurons. Thus, it seems likely that significant Kv2/KvS heteromeric conductances also exist in specific subsets of brain neurons.

The physiological role of Kv2/KvS heteromers in neurons and other excitable cells remains enigmatic. Previous studies have used knockout mice (Regnier et al., 2017; Miyamae et al., 2021), transient transfection of KvS subunits (Lee et al., 2020b), siRNA knockdown (Tsantoulas et al., 2012), and modeling (Miyamae et al., 2021) to probe the presence of endogenous and functional KvS-containing channels, and these methods can identify phenotypic changes that suggest potential roles of KvS-containing channels. Application of GxTX *in vivo* could be used to probe the physiological roles of Kv2 with or without KvS subunits, while RY785 selectively targets Kv2-only channels. Genetic mutations and gene targeting studies have linked disruptions in the function of KvS-containing channels to epilepsy (Jorge et al., 2011), neuropathic pain sensitivity (Tsantoulas et al., 2018), labor pain (Lee et al., 2020a) and retinal cone dystrophy (Wu et al., 2006; Hart et al., 2019; Inamdar et al., 2022), stressing their functional importance in specific cell types. The unique pharmacology of KvS-containing channels identified here provides a new and direct method of identifying conductances mediated by KvS-containing channels in native neurons and establishing what contributions KvS subunits make to electrophysiological signaling.

Finally, these findings also support the potential utility of KvS channels as drug targets. Kv2-targeted drug leads have poor tissue and cell specificity and suffer from pronounced side effects (Li et al., 2013). KvS transcripts show far greater tissue- and cell-type specific expression relative to Kv2 (Bishop et al., 2015; Bocksteins, 2016), and we identified prominent Kv2 conductances with KvS-like pharmacology in mouse nociceptor and human DRG neurons. Consequently, KvS-targeted drugs could offer greater specificity and the ability to modulate neuronal excitability in a variety of pathological contexts, such as neuropathic pain.

Materials and methods

Mice

Studies were approved by the UC Davis and Harvard Medical School Institutional Animal Care and Use Committees and conform to guidelines established by the NIH. Mice were maintained on a 12 hr light/dark cycle, and food and water were provided *ad libitum*. The *Mrgprd^{GFP}* (MGI: 3521853) and *CGRP^{GFP}* (MGI: 2151253) mouse lines were a generous gift from David Ginty at Harvard.

Human tissue collection

Human dorsal root ganglia (DRG) were obtained from Sierra Donor Services. The donor was a 58-year-old Asian Indian female and DRG were from the 1st and 2nd lumbar region (cause of death: Stroke). DRG were extracted 6 hours after aortic cross clamp and placed in an ice cold N-methyl-D-glucamine-artificial cerebral spinal fluid (NMDG-aCSF) solution containing in mM: 93 NMDG, 2.5 KCl, 1.25 NaH₂PO₄, 30 NaHCO₃, 20 HEPES, 25 Glucose, 5 L-Ascorbic acid, 2 Thiourea, 3 Na pyruvate, 10 MgSO₄ and 0.5 CaCl₂ pH adjusted to 7.4 with HCl. Human DRG were obtained from the organ donor with full legal consent for use of tissue for research in compliance with procedures approved by Sierra Donor Services.

Chinese hamster ovary (CHO) cell culture and transfection

The CHO-K1 cell line transfected with a tetracycline-inducible rat Kv2.1 construct (Kv2.1-CHO) (Trapani and Korn, 2003) was cultured as described previously (Tilley et al., 2014). Transfections were achieved with Lipofectamine 3000 (Life Technologies, L3000001). 1 μ l Lipofectamine was diluted, mixed, and incubated in 25 μ l of Opti-MEM (Gibco, 31985062). Concurrently, 0.5 μ g of KvS or AMIGO1 or Nav β 2, 0.5 μ g of pEGFP, 2 μ l of P3000 reagent and 25 μ l of Opti-MEM were mixed. DNA and Lipofectamine 3000 mixtures were mixed and incubated at room temperature for 15 min. This transfection cocktail was added to 1 ml of culture media in a 24 well cell culture dish containing Kv2.1-CHO cells and incubated at 37 °C in 5% CO₂ for 6 h before the media was replaced. Immediately after media was replaced, Kv2.1 expression was induced in Kv2.1-CHO cells with 1 μ g/ml minocycline (Enzo Life Sciences, ALX-380-109-M050), prepared in 70% ethanol at 2 mg/ml. Voltage clamp recordings were performed 12-24 hours later. We note that the expression method of Kv2/KvS heteromers used here is distinct from previous studies which show that the KvS:Kv2 mRNA ratio can affect the expression of functional Kv2/KvS heteromers (Salinas et al., 1997b; Pisupati et al., 2018). We validated the functional Kv2/KvS heteromer expression using voltage clamp to establish distinct channel kinetics and the presence of RY785-resistant conductance in KvS-transfected cells and using immunohistochemistry to label apparent surface localization of KvS subunits (Figure 4, Figure 1 Supplement, Figure 1 and Figure 5). During recordings, the experimenter was blinded as to whether cells had been transfected with KvS, or Nav β 2 or AMIGO1. Human Kv5.1, human Kv6.4 and human Kv8.1, AMIGO1-YFP, and pEGFP plasmids were gifts from James Trimmer (University of California, Davis, CA). Human Kv9.1 and human Kv9.3 plasmids were purchased from Addgene. Human Nav β 2 plasmid was a kind gift from Dr. Alfred George (Lossin et al., 2002).

Neuron cell culture

Mouse

Cervical, thoracic and lumbar dorsal root ganglia (DRGs) were harvested from 7- to 10-week-old *Mrgprd*^{GFP} or 2- to 4-week-old *CGRP*^{GFP} mice and transferred to Hank's buffered saline solution (HBSS) (Invitrogen). Ganglia were treated with collagenase (2 mg/ml; Type P, Sigma-Aldrich) in HBSS for 15 min at 37 °C followed by 0.05% Trypsin-EDTA (Gibco) for 2.5 min with gentle rotation. Trypsin was neutralized with culture media (MEM, with l-glutamine, Phenol Red, without sodium pyruvate) supplemented with 10% horse serum (heat-inactivated; Gibco), 10 U/ml penicillin, 10 μ g/ml streptomycin, MEM vitamin solution (Gibco), and B-27 supplement (Gibco). Serum-containing media was decanted and cells were triturated using a fire-polished Pasteur pipette in MEM culture media containing the supplements listed above. Cells were plated on laminin-treated (0.05 mg/ml, Sigma-Aldrich) 5 mm German glass coverslips (Bellco Glass, 1943-00005), which had previously been washed in 70% ethanol and sterilized with ultraviolet light. Cells were then incubated at 37 °C in 5% CO₂. Cells were used for electrophysiological experiments 24–38 hr after plating.

Superior cervical ganglia (SCG) were harvested from Swiss Webster (CFW) mice (postnatal day 13–15, either sex) and treated for 20 min at room temperature (RT) with 20 U/ml papain (Worthington Biochemical), 5 mM dl-cysteine, 1.25 mM EDTA, and 67 μ M β -mercaptoethanol in a Ca²⁺, Mg²⁺-free (CMF) Hank's solution (Gibco) supplemented with 1 mM Sodium Pyruvate (Sigma-Aldrich, St. Louis, MO), and 5 mM HEPES (Sigma-Aldrich, St. Louis, MO). Ganglia were then treated for 20 min at 37 °C with 3 mg/ml collagenase (type I; Roche Diagnostics) and 3 mg/ml dispase II (Roche Diagnostics) in CMF Hank's solution. Cells were dispersed by trituration with fire-polished Pasteur pipettes in a solution composed of two media combined in a 1:1 ratio: Leibovitz's L-15 (Invitrogen) supplemented with 5 mM HEPES, and DMEM/F12 medium (Invitrogen). Cells were then plated on

glass coverslips and incubated at 37 °C (95% O₂, 5% CO₂) for 1 hr, after which Neurobasal medium (Invitrogen) with B-27 supplement (Invitrogen), penicillin and streptomycin (Sigma) was added to the dish. Cells were incubated at 25 °C (95% O₂, 5% CO₂) and used within 10 hr.

Human

Dura were removed from human DRG with a scalpel in ice cold NMDG-aCSF solution (Valtcheva et al., 2016). Human DRG were then cut into approximately 1 mm thick sections and were placed in 1.7 mg/mL Stemxyme (Worthington Biochemical, LS004107) and 6.7 mg/mL DNase I (Worthington Biochemical, LSO02139) diluted in HBSS (Thermo Fisher Scientific, 14170161) for 12 hr at 37 °C. DRG were then triturated with a fire-polished Pasteur pipette and passed through a 100 µm cell strainer. Cells were then spun at 900 g through 10% BSA. The supernatant was removed, and cells were resuspended in human DRG culturing media that contained 1% penicillin/streptomycin, 1% GlutaMAX (Gibco, 35050-061), 2% NeuroCult SM1 (05711, Stemcell technologies), 1% N2 Supplement (Thermo Scientific, 17502048), 2% FBS (Gibco, 26140-079) diluted in BrainPhys media (Stemcell technologies, 05790). DRG neurons were plated on poly-D-lysine treated (0.01 mg/mL) 5 mm German glass coverslips, which had previously been washed in 70% ethanol and sterilized with ultraviolet light. DRG neurons were then incubated at 37 °C in 5% CO₂. Human DRG neuron experiments were performed up to 7 days after plating.

Whole cell voltage clamp of CHO cells

Voltage clamp was achieved with a dPatch amplifier (Sutter Instruments) run by SutterPatch software (Sutter Instruments). Solutions for Kv2.1-CHO cell voltage-clamp recordings: CHO-internal (in mM) 120 K-methylsulfonate, 10 KCl, 10 NaCl, 5 EGTA, 0.5 CaCl₂, 10 HEPES, 2.5 MgATP pH adjusted to 7.2 with KOH, 289 mOsm. CHO-external (in mM) 145 NaCl, 5 KCl, 2 CaCl₂, 2 MgCl₂, 10 HEPES pH adjusted to 7.3 with NaOH, 298 mOsm. Osmolality was measured with a vapor pressure osmometer (Wescor, 5520). The liquid junction potential of -9 mV between these solutions was accounted for. The liquid junction potential was calculated according to the stationary Nernst-Planck equation (Marino et al., 2014) using LJPCalc (RRID:SCR_025044). For voltage-clamp recordings, Kv2.1-CHO cells were detached in a PBS-EDTA solution (Gibco, 15040-066), spun at 500 g for 2 minutes and then resuspended in 50% cell culture media and 50% CHO-external recording solution. Cells were then added to a recording chamber (Warner, 64-0381) and were rinsed with the CHO-external patching solution after adhering to the bottom of the recording chamber. Transfected Kv2.1-CHO cells were identified by GFP fluorescence and were selected for whole cell voltage clamp. Thin-wall borosilicate glass recording pipettes (Sutter, BF150-110-10) were pulled with blunt tips, coated with silicone elastomer (Sylgard 184, Dow Corning), heat cured, and tip fire-polished to resistances less than 4 MΩ. Series resistance of 2–14 MΩ was estimated from the Sutterpatch whole-cell parameters routine. Series resistance compensation between 13 and 90% was used to constrain voltage error to less than 15 mV; compensation feedback lag was 6 µs for most experiments or 100 µs for concentration-effect experiments. Capacitance and ohmic leak were subtracted using a P/4 protocol. Output was low-pass filtered at 5 kHz using the amplifier's built-in Bessel and digitized at 25 kHz or, for concentration-effect experiments, 1 and 10 kHz. Experiments were performed on Kv2.1-CHO cells with membrane resistance greater than 1 GΩ assessed prior to running voltage clamp protocols while cells were held at a membrane potential of -89 mV. RY785 (gift from Bruce Bean, Harvard, or Cayman, 19813) was prepared in DMSO as a 1 mM stock for dilutions to 1 µM or a 35 mM stock for concentration-effect experiments. Stocks of GxTX-1E Met35Nle (Tilley et al., 2014) in water were 10 µM. Stocks were stored frozen and diluted in recording solution just prior to application to cells. Solutions were flushed over cells at a rate of approximately 1 mL/min. Concentrated RY785 and GxTX stocks were stored at -20 °C. Kv2.1-CHO cells were given voltage steps from -89 mV to -9 mV for 200 ms every 6 seconds during application of RY785 until currents stabilized. When vehicle control was applied to cells, -9 mV steps were given for a similar duration. All RY785 solutions contained 0.1% DMSO. Vehicle control solutions also contained 0.1% DMSO but lacked RY785. Perfusion lines were cleaned with 70% ethanol then doubly-deionized water. For concentration-effect experiments, changes in current

amplitude due to solution exchange were controlled for by treating every other tested cell with multiple washes of the same, 0.35 μM RY785 solution instead of increasing concentrations of RY785. The timing and duration of these control washes was similar to that of the washes in concentration-effect experiments.

Whole cell voltage clamp of mouse and human dorsal root ganglion neurons

Whole cell recordings from *Mrgprd*^{GFP} mouse and human neurons were performed using the same methods as CHO cell recordings with the following exceptions. Voltage clamp was achieved with a dPatch amplifier run by SutterPatch software or an AxoPatch 200B amplifier (Molecular Devices) controlled by PatchMaster software (v2×91, HEKA Elektronik) via an ITC-18 A/D board (HEKA Instruments Inc). Solutions for voltage-clamp recordings: internal (in mM) 140 KCl, 13.5 NaCl, 1.8 MgCl₂, 0.09 EGTA, 4 MgATP, 0.3 Na₂GTP, 9 HEPES pH adjusted to 7.2 with KOH, 326 mOsm. The external solution contained (in mM) 3.5 KCl, 155 NaCl, 1 MgCl₂, 1.5 CaCl₂, 0.01 CdCl₂, 10 HEPES, 10 glucose pH adjusted to 7.4 with NaOH, 325 mOsm. The calculated liquid junction potential of -4 mV between these solutions was accounted for. For voltage-clamp recordings, neurons on cover slips were placed in the same recording chamber used for CHO cell recordings and were rinsed with an external patching solution. Neurons from *Mrgprd*^{GFP} mice with green fluorescence were selected for recordings. Human DRG neurons with cell capacitances between 22.5 and 60 pF were used. After whole-cell voltage clamp was established, non-Kv2/KvS conductances were suppressed by changing to an external solution containing a cocktail of inhibitors: 100 nM alpha-dendrotoxin (Alomone) to block Kv1 (Harvey and Robertson, 2004 [\[link\]](#)), 3 μM AmmTX3 (Alomone) to block Kv4 (Maffie et al., 2013 [\[link\]](#); Pathak et al., 2016 [\[link\]](#)), 100 μM 4-aminopyridine to block Kv3 (Coetzee et al., 1999 [\[link\]](#); Gutman et al., 2005 [\[link\]](#)), 1 μM TTX to block TTX sensitive Nav channels, and 10 μM A-803467 (Tocris) to block Nav1.8 (Jarvis et al., 2007 [\[link\]](#)). It is possible that off target effects of blockers may introduce errors in the quantification Kv2/KvS heteromer-mediated K⁺ currents. For example, 4-aminopyridine is expected to block a small fraction, 2%, of Kv2 homomers and have a lesser impact on Kv2/KvS heteromers (Post et al., 1996 [\[link\]](#); Thorneloe and Nelson, 2003 [\[link\]](#); Stas et al., 2015 [\[link\]](#)) which could result in a slight overestimation of the ratio of Kv2/KvS heteromers to Kv2 homomers. After addition of 1 μM RY785, neurons were given 10 steps to -24 mV for 500 ms to allow for voltage dependent block of RY785. Thin-wall borosilicate glass recording pipettes were pulled with blunt tips, coated with silicone elastomer, heat cured, and tip fire-polished to resistances less than 2 M Ω . Series resistance of 1–4 M Ω was estimated from the whole-cell parameters circuit. Series resistance compensation between 55 and 98% was used to constrain voltage error to less than 15 mV. Ohmic leak was not subtracted. Neurons were held at a membrane potential of -74 mV to mimic a physiological resting potential. KvS subunits can profoundly shift the voltage-inactivation relation (Salinas et al., 1997a [\[link\]](#); Kramer et al., 1998 [\[link\]](#); Kerschensteiner and Stocker, 1999 [\[link\]](#)) and this potential is likely insufficiently negative to relieve inactivation from all Kv2/KvS heteromeric channels. Also, the activation membrane potential is close to the half-maximal point of Kv2/KvS conductances. Thus the ratio of Kv2-like to KvS-like conductance is expected to vary with voltage protocols.

Whole cell recordings from *CGRP*^{GFP} mice were made using the same methods as *Mrgprd*^{GFP} mice with the following exceptions. The internal solution contained in mM: 139.5 KGluconate, 1.6 MgCl₂, 1 EGTA, 0.09 CaCl₂, 9 HEPES, 14 creatine phosphate, 4 MgATP, 0.3 GTP pH adjusted to 7.2 with KOH. The external solution contained in mM: 155 NaCl, 3.5 KCl, 1 MgCl₂, 1.5 CaCl₂, 10 glucose, 10 HEPES pH adjusted to 7.4 with NaOH and did not contain the cocktail of inhibitors used for recordings with *Mrgprd*^{GFP} mice. The liquid junction potential of -13 mV between these solutions was accounted for. Ohmic leak was subtracted. After establishing whole-cell recording, cells were lifted and placed in front of a series of quartz fiber flow pipes for rapid solution exchange and application of RY785 and GxTX.

Whole cell recordings from mouse superior cervical ganglion neurons were performed using an Axon Instruments Multiclamp 700B Amplifier (Molecular Devices). Electrodes were pulled on a Sutter P-97 puller (Sutter Instruments) and shanks were wrapped with Parafilm (American National Can Company) to allow optimal series resistance compensation without oscillation. Voltage or current commands were delivered and signals were recorded using a Digidata 1321A data acquisition system (Molecular Devices) controlled by pCLAMP 9.2 software (Molecular Devices). The internal solution was (in mM): 140 mM K aspartate, 13.5 mM NaCl, 1.8 mM MgCl₂, 0.09 mM EGTA, 9 mM HEPES, 14 mM creatine phosphate (Tris salt), 4 mM MgATP, 0.3 mM Tris-GTP, pH 7.2 adjusted with KOH. The base external solution was the same as for DRG recordings. The calculated liquid junction potential of -15 mV between these solutions was accounted for. After establishing whole-cell recording, the cell was lifted and placed in front of a series of quartz fiber flow pipes for rapid solution exchange and application of RY785 and GxTX. The external solution used for recording Kv2 currents used the same cocktail of inhibitors for sodium channels and other potassium channels as for the DRG recordings except that A-803467 was omitted because the sodium current in SCG neurons is all TTX sensitive (Toledo-Aral et al., 1997 [\[link\]](#)).

Voltage clamp analysis

Activation kinetics were fit from 10-90% of current (I_K) rise with the power of an exponential function:

Equation 1

$$I_K = A \left(1 - e^{-\frac{t}{\tau_{act}}} \right)^\sigma$$

where A is the maximum current amplitude, τ_{act} is the time constant of activation, σ is sigmoidicity, and t is time. The $t = 0$ mark was adjusted to 100 μ s after the start of the voltage step from the holding potential to correct for filter delay and cell charging.

Conductance values were determined from tail current levels at -9 mV after 200 ms steps to the indicated voltage. Tail currents were the mean current amplitude from 1 to 5 ms into the -9 mV step. Conductance–voltage relations were fit with the Boltzmann function:

Equation 2

$$f(V) = A \left(1 + e^{-\left(\frac{V - V_{1/2}}{RT} \right) \frac{zF}{RT}} \right)^{-1}$$

where V is voltage, A is amplitude, z is the number of elementary charges, F is Faraday's constant, R is the universal gas constant, and T is temperature (held at 295 K).

Deactivation kinetics were fit with a double exponential:

Equation 3

$$f(x) = y_0 + A_1 e^{-\frac{(t-t_0)}{\tau_1}} + A_2 e^{-\frac{(t-t_0)}{\tau_2}}$$

Where t is time, y_0 is the initial current amplitude, t_0 is the start time of the exponential decay, τ_1 and τ_2 are the time constants, and A_1 and A_2 are the amplitudes of each component.

In CHO cells peak currents were analyzed because outward currents seem to offer the best signal/noise. In neurons, we restricted analysis to tail currents at elapsed times to minimize complications from non-Kv2 endogenous voltage-gated channels which deactivate more quickly. In neurons, voltage gated currents remained in the toxin cocktail + RY785 and GxTX, that were sometimes unstable. To minimize complications from these currents, we restricted analysis of RY785 and GxTX subtraction experiments to tail currents at elapsed times to minimize complications from non-Kv2 endogenous voltage-gated channels which deactivate more quickly.

We note that the analysis of conductance activation by using tail currents is only accurate when dealing with non-inactivating conductances. We expect that inactivation of Kv2/KvS conductances during the 200 ms pre-pulse is minimal (Salinas et al., 1997a [↗](#); Kramer et al., 1998 [↗](#); Kerschensteiner and Stocker, 1999 [↗](#)) and did not notice inactivation during the activation pulse. Also, deactivation kinetics can vary in a heterogenous population of Kv2/KvS heteromers. While analysis of tail currents could skew the quantification of total Kv2 like and KvS-like conductances, our data supports that mouse nociceptors and human neurons have tail currents that are resistant to RY785 and sensitive to GxTX consistent with the presence of Kv2/KvS heteromers.

Preparation of rKv2.1 and hKv2.1/hKv8.1 docking inputs

The rKv2.1 homomer (PDB ID: 8SD3; Fernández-Mariño et al., 2023) was energy minimized using cryo-EM structure refinement with Rosetta (Wang et al., 2016). This energy-minimized model was used for generation of the hKv2.1/hKv8.1 heteromer using ColabFold (Mirdita et al., 2022) with 48 recycles, dropout enabled, and Amber minimization (Salomon-Ferrer et al., 2013). To provide the same pore conformation as input between homomer and heteromer docking, the Kv8.1 subunit from ColabFold was superimposed to a Kv2.1 subunit from the energy-minimized model used for homomer docking; this heteromer model was then energy minimized by fixing the Kv2.1 backbone and sidechain chi torsion angles while allowing the Kv8.1 backbone and sidechain chi torsion angles to minimize with Rosetta FastRelax (Tyka et al., 2011) using the BetaNov16 scoring function (Pavlovicz et al., 2020).

Preparation of RY785 docking input

RY785 was extracted as a structure data file from PubChem (Kim et al., 2023). With Avogadro (Hanwell et al., 2012), RY785 underwent bond correction, protonation at pH 7.4, and energy minimization using the Merck molecular force field (Halgren, 1996a; b; c; Halgren and Nachbar, 1996; Halgren, 1999a; b). Next, using the Antechamber protocol of AmberTools (Case et al., 2023), the partial atomic charge, atom, and bond-type assignments for each ligand were AM1-BCC corrected. The input conformer was generated using the RosettaGenFF (Park et al., 2021) crystal structure prediction protocol, taking the lowest energy packing arrangement as input.

RY785 Docking

Docking was performed using Rosetta's GALigandDock (Park et al., 2021) for ion channel docking as described previously (Harris et al., 2024). Briefly, GALigandDock in the flexible docking mode was used with a padding value of 7 Å, 20 generations with a pool of 100 poses, and the entire pool of poses as output. With this run mode, the entire pore cavity had the potential to be sampled with side chain flexibility. We ran GALigandDock 100 times, generating 10,000 total models for both the rKv2.1 homomer and hKv2.1/hKv8.1 heteromer. The top ten models were selected for visual inspection by sorting the top 10% of models by total score, followed with the top 10 of the subset by interface score. Contact analysis was performed using UCSF ChimeraX (Pettersen et al., 2021) Clashes and Contacts tool and the H-Bonds tool with default distance cutoff and acceptance settings.

Immunofluorescence

Kv2.1-CHO cells were fixed for 15 minutes at 4 °C in 4% formaldehyde prepared fresh from paraformaldehyde in PBS buffer pH 7.4. Cells were then washed 3 x 5 minutes in PBS, followed by blocking in blotto-PBS (PBS, pH 7.4 with 4% (w/v) non-fat milk powder and 0.1% (v/v) Triton-X100) for 1 hr. Cells were incubated for 1 hr with primary antibodies diluted in blotto-PBS and subsequently washed 3 x 5 minutes in PBS. Antibodies used were mAb K89/34 for Kv2.1 (NeuroMab, RRID:AB_1067225), rabbit pAb 5.1C for Kv5.1 (in-house, RRID:AB_3076240), and rabbit anti-V5 for Kv9.3-V5 (Rockland, 600-401-378). For surface labeling of Kv5.1, non-permeabilized cells were incubated with Kv5.1 mAb (Santa Cruz Biotech, 81881) in blotto-PBS lacking Triton-X100. The cells were then incubated with mouse IgG subclass- and/or species-specific Alexa-

conjugated fluorescent secondary antibodies (Invitrogen) diluted in blotto-PBS for 45 min and washed 3 x 5 min in PBS. Cover glasses were mounted on microscope slides with Prolong Gold mounting medium (ThermoFisher, P36930) according to the manufacturer's instructions. Widefield fluorescence images were acquired with an AxioCam MRm digital camera installed on a Zeiss AxioImager M2 microscope with a 63x/1.40 NA Plan-Apochromat oil immersion objective and an ApoTome coupled to Axiovision software version 4.8.2.0 (Zeiss, Oberkochen, Germany).

Multiplex in situ hybridization

A 6-week-old *Mrgprd*^{GFP} mouse was briefly anesthetized with 3–5% isoflurane and then decapitated. The spinal column was dissected, and the left and right L1 DRG were removed and drop fixed for 12 minutes in ice cold 4% paraformaldehyde in 0.1 M phosphate buffer (PB) pH adjusted to 7.4. The L1 vertebrae was identified by the 13th rib. The DRG was washed 3 × 10 minutes each in PB and cryoprotected at 4 °C in 30% sucrose diluted in PB for 2 hr. The DRG were then frozen in Optimal Cutting Temperature (OCT) compound (Fisher, 4585) and stored at –80 °C until sectioning. Samples were cut into 20 µm sections on a freezing stage sliding microtome and were collected on Colorfrost Plus microscope slides (Fisher Scientific, 12-550-19). Sections were processed for RNA *in situ* detection using an RNAscope Fluorescent Detection Kit according to the manufacturer's instructions (Advanced Cell Diagnostics) with the following probes: *KCNF1* (508731, mouse) or *KCNS1* (525941, mouse). TSA Vivid 650 Fluorophore was used to label probes (TSA Vivid, 7527) Following in situ hybridization, immunohistochemistry to label GFP was performed. Sections were incubated in vehicle solution (4% milk, 0.2% triton diluted in PB) for 1 hr at RT. Tissue was then incubated in a rabbit polyclonal anti-GFP antibody (Rockland 600-401-215S) diluted 1:1000 in vehicle overnight at 4 °C. Sections were washed three times in vehicle for 5 minutes per wash and then incubated in a goat anti-rabbit secondary antibody (Invitrogen, A-11008) diluted 1:1500 in vehicle. Sections were then mounted with Prolong Gold (Thermo Fisher, P36930) and #1.5 cover glass (Fisher Scientific, NC1776158).

Imaging

Images were acquired with an inverted scanning confocal and airy disk imaging system (Zeiss LSM 880 Airyscan, 410900-247-075) run by ZEN black v2.1. Laser lines were 488 nm and 633 nm. Images were acquired with a 0.8 NA 20x objective (Zeiss, 420650-9901) details in figure legends.

Statistics

All statistical tests were performed in Igor Pro software version 8 (Wavemetrics, Lake Oswego, OR). Independent replicates (n) are individual cells/neurons while biological replicates (N) are individual mice. All tests were two-tailed. Wilcoxon rank tests were used for two-sample comparisons. Dunnett tests were used for multiple comparisons.

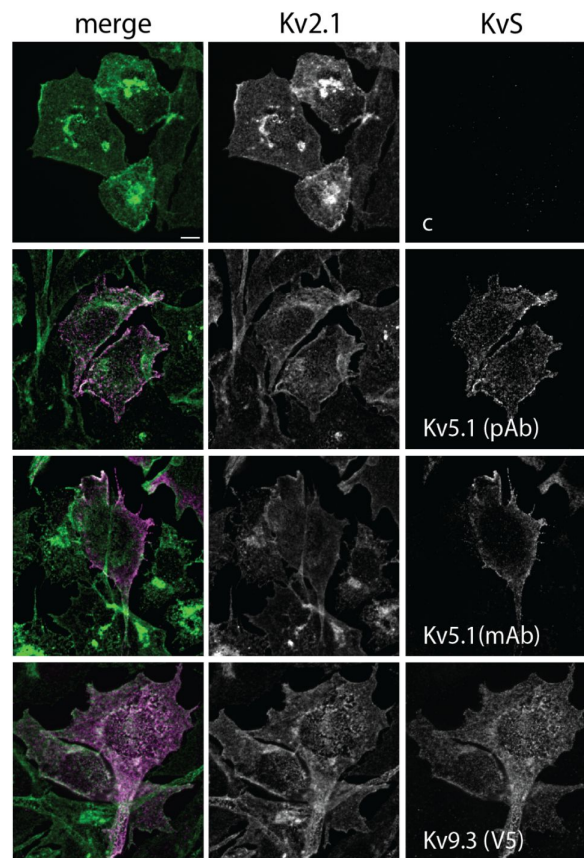


Figure 1 Supplement.

KvS subunits colocalize with Kv2.1 on the surface of CHO cells.

Kv2.1-CHO cells transfected with the designated KvS subunits (c = no KvS transfection) were immunolabeled for Kv2.1 (green) and Kv5.1 or Kv9.3 (magenta). Immunolabeling for Kv5.1 and Kv9.3 were detected both intracellularly and on the apparent cell surface where they colocalized with Kv2.1 labeling. The anti-Kv5.1 mAb recognizes an extracellular epitope and was used on non-permeabilized cells, confirming surface expression of Kv5.1. Scale bars = 5 μ m.

Figure 2 Supplement.

Rapid unblock of RY785 from Kv2.1/Kv8.1 heteromers

A) Exemplar traces showing current recovery following solution exchange from 35 μM RY785 into 0.35 μM RY785. Voltage protocol is the same as in [Figure 2](#). Cells were held at -89 mV without pulsing during solution exchange.

B) Tail current amplitudes, as in [Figure 2](#), normalized to pulse #5. $n = 5$ cells treated first with 35 μM RY785 (dark red) then 0.35 μM RY785 (orange). $n = 4$ control cells treated first with 0.35 μM RY785 and washed again with 0.35 μM RY785 (pink). The transient increase in current amplitudes upon solution exchange also occurs with Kv2.1 ([Marquis and Sack, 2022](#)).

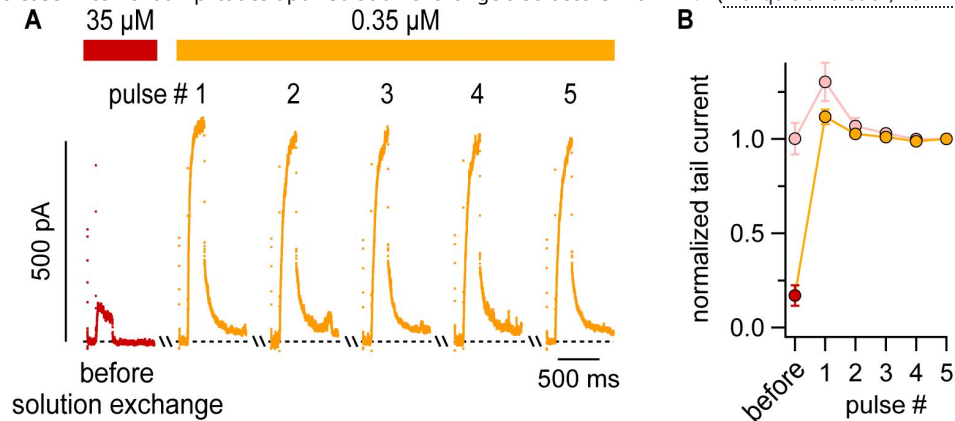


Figure 4 Supplement.

Effect of vehicle control on Kv2.1.

Left: Exemplar traces from a Kv2.1-CHO cell transfected with Nav β 2. Black and green traces are currents before and after application of vehicle control respectively. Right: Current remaining after application of vehicle control. Black bar represents mean. Each point represents current from one cell at the end of a 200 ms -9 mV voltage step.

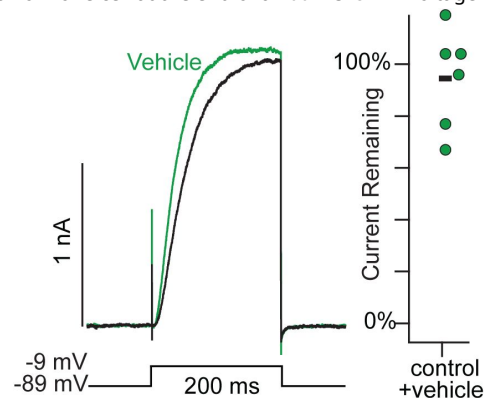


Figure 7 Supplement 1.

Nonpeptidergic nociceptors express Kv5.1 and Kv9.1 mRNA transcripts.

Exemplar images of DRG sections from a *Mrgprd*^{GFP} mouse labeled with RNAscope *in situ* hybridization for *KCNF1* (Kv5.1) (top magenta) or *KCNS1* (Kv9.1) (bottom magenta). Scale bars are 50 μ m.

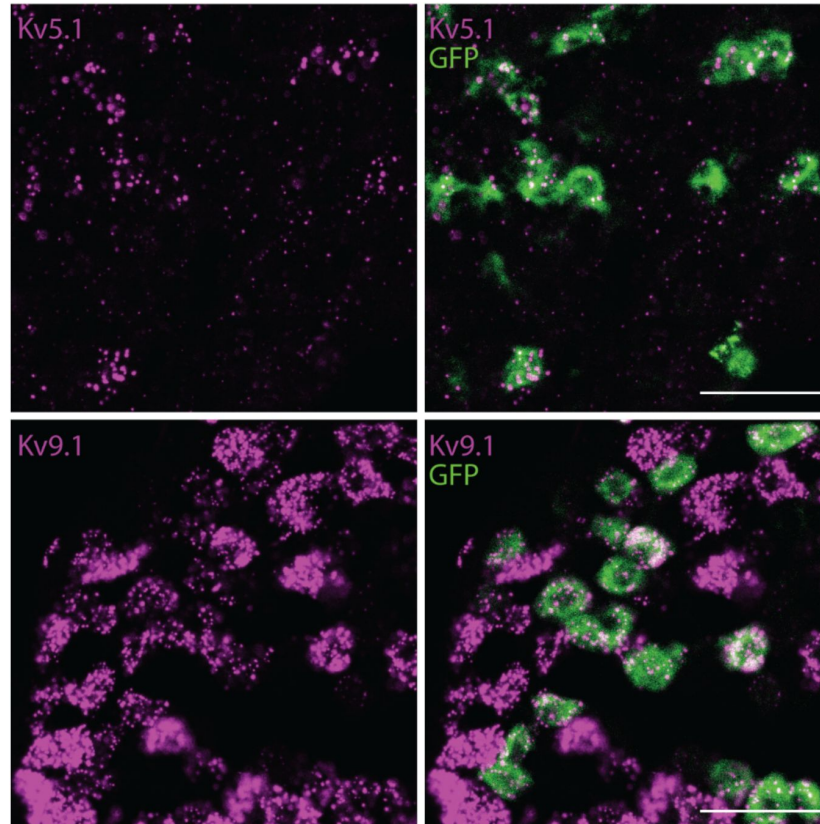
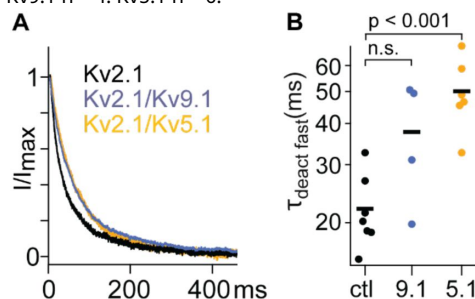


Figure 7 Supplement 2.

RY785 resistant currents from Kv2.1-CHO cells transfected with Kv5.1 or Kv9.1 deactivate slower than currents from untransfected Kv2.1-CHO cell.

A) Exemplar traces of channel deactivation at -49 mV after a 50 ms step to +11 mV. Traces are normalized to max current during the -49 mV step.

B) The faster time constant of a double exponential fit (Equation 3) to channel deactivation. Dunnett tests versus control (ctl). Bars represent means. Ctl n = 7. Kv9.1 n = 4. Kv5.1 n = 6.



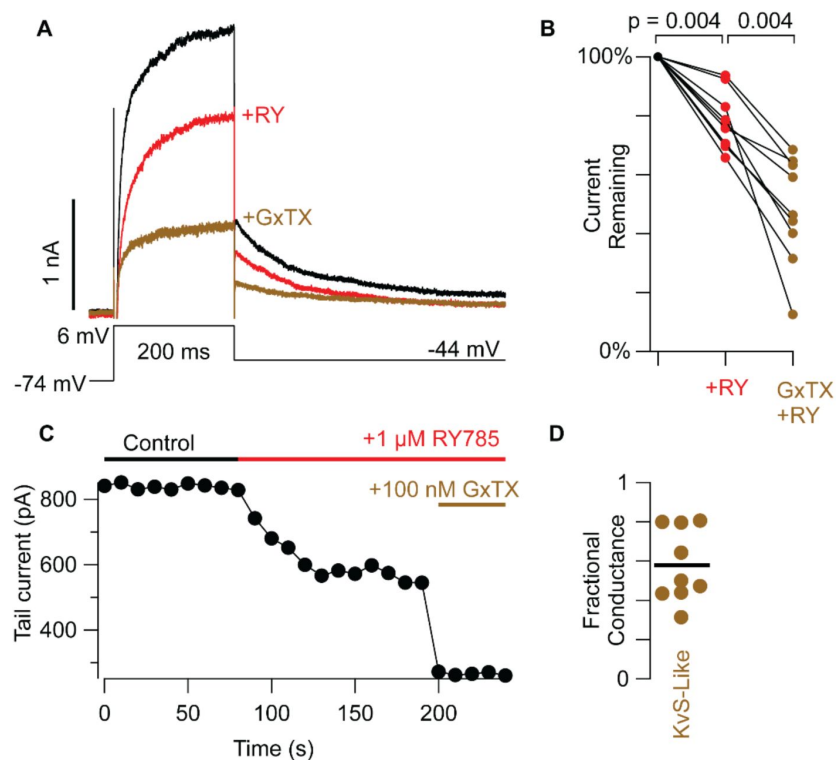


Figure 7 Supplement 3

The Kv2 conductances of mouse dorsal root ganglion neurons have KvS-like pharmacology in the absence of the cocktail of inhibitors.

- A) Exemplar currents from peptidergic nociceptor, GFP⁺ neurons from *CGRP^{GFP}* mice. Inward sodium currents are cropped so that only outward currents are shown.
- B) Tail current amplitude 10 ms after voltage was stepped from +6 mV to -44 mV normalized to current amplitude before RY785. Wilcoxon rank tests were paired. RY785 then GxTX: n = 9 neurons, N = 3 mice.
- C) Time course of inhibition of tail currents in A in control and during drug application.
- D) Fractional KvS-like conductance relative to the total RY785 + GxTX-sensitive conductance. KvS-like is only sensitive to GxTX. Bar represents mean.

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References

- Aras MA, Saadi RA, Aizenman E (2009) **Zn²⁺ regulates Kv2.1 voltage-dependent gating and localization following ischemia** *Eur J Neurosci* **30**:2250–2257
- Bishop HI, Guan D, Bocksteins E, Parajuli LK, Murray KD, Cobb MM, Misonou H, Zito K, Foehring RC, Trimmer JS (2015) **Distinct cell- and layer-specific expression patterns and independent regulation of Kv2 channel subtypes in cortical pyramidal neurons** *J Neurosci* **35**:14922–14942
- Bocksteins E (2016) **Kv5, Kv6, Kv8, and Kv9 subunits: No simple silent bystanders** *J Gen Physiol* **147**:105–125
- Bocksteins E, Snyders DJ (2012) **Electrically silent Kv subunits: their molecular and functional characteristics** *Physiology* **27**:73–84
- Bocksteins E, Snyders DJ, Holmgren M (2017) **Independent movement of the voltage sensors in K(V)2.1/K(V)6.4 heterotetramers** *Sci Rep* **7**:41646
- Bocksteins E, Labro AJ, Snyders DJ, Mohapatra DP (2012) **The electrically silent Kv6.4 subunit confers hyperpolarized gating charge movement in Kv2.1/Kv6.4 heterotetrameric channels** *PLoS One* **7**:e37143
- Bocksteins E, Raes AL, Van de Vijver G, Bruyns T, Van Bogaert PP, Snyders DJ (2009) **Kv2.1 and silent Kv subunits underlie the delayed rectifier K⁺ current in cultured small mouse DRG neurons** *Am J Physiol Cell Physiol* **296**:C1271–1278
- Castellano A, Chiara MD, Mellstrom B, Molina A, Monje F, Naranjo JR, Lopez-Barneo J (1997) **Identification and functional characterization of a K⁺ channel alpha-subunit with regulatory properties specific to brain** *J Neurosci* **17**:4652–4661
- Coetzee WA, Amarillo Y, Chiu J, Chow A, Lau D, McCormack T, Moreno H, Nadal MS, Ozaita A, Pountney D, Saganich M, Vega-Saenz de Miera E, Rudy B (1999) **Molecular diversity of K⁺ channels** *Ann N Y Acad Sci* **868**:233–285
- Du J, Haak LL, Phillips-Tansey E, Russell JT, McBain CJ (2000) **Frequency-dependent regulation of rat hippocampal somato-dendritic excitability by the K⁺ channel subunit Kv2.1** *J Physiol* **522**:19–31
- Ferns M, van der List D, Vierra NC, Lacey T, Murray K, Kirmiz M, Stewart RG, Sack JT, Trimmer JS (2025) **The electrically silent Kv5.1 subunit forms heteromeric Kv2 channels in cortical neurons and confers distinct functional properties** *J Neurosci*
- Gong S, Zheng C, Doughty ML, Losos K, Didkovsky N, Schambra UB, Nowak NJ, Joyner A, Leblanc G, Hatten ME, Heintz N (2003) **A gene expression atlas of the central nervous system based on bacterial artificial chromosomes** *Nature* **425**:917–925
- Gutman GA, Chandy KG, Grissmer S, Lazdunski M, McKinnon D, Pardo LA, Robertson GA, Rudy B, Sanguinetti MC, Stühmer W, Wang X (2005) **International Union of Pharmacology LIII.**

Nomenclature and molecular relationships of voltage-gated potassium channels. Pharmacol Rev **57**:473–508

Hart NS, Mountford JK, Voigt V, Fuller-Carter P, Barth M, Nerbonne JM, Hunt DM, Carvalho LS (2019) **The Role of the Voltage-Gated Potassium Channel Proteins Kv8.2 and Kv2.1 in Vision and Retinal Disease: Insights from the Study of Mouse Gene Knock-Out Mutations** *eNeuro* **6**:ENEURO.0032–19

Harvey AL, Robertson B (2004) **Dendrotoxins: structure-activity relationships and effects on potassium ion channels** *Curr Med Chem* **11**:3065–3072

Herrington J, Solly K, Ratliff KS, Li N, Zhou YP, Howard A, Kiss L, Garcia ML, McManus OB, Deng Q, Desai R, Xiong Y, Kaczorowski GJ (2011) **Identification of novel and selective Kv2 channel inhibitors** *Mol Pharmacol* **80**:959–964

Herrington J, et al. (2006) **Blockers of the delayed-rectifier potassium current in pancreatic beta-cells enhance glucose-dependent insulin secretion** *Diabetes* **55**:1034–1042

Huang X, Lee S, Chen K, Kawaguchi R, Wiskow O, Ghosh S, Frost D, Perrault L, Pandey R, Klim JR, Boivin B, Hermawan C, Livak KJ, Geschwind DH, Wainger BJ, Eggan KC, Bean BP, Woolf CJ (2024) **Downregulation of the silent potassium channel Kv8.1 increases motor neuron vulnerability in amyotrophic lateral sclerosis** *Brain Commun* **6**:fcae202

Inamdar SM, Lankford CK, Poria D, Laird JG, Solessio E, Kefalov VJ, Baker SA (2022) **Differential impact of Kv8.2 loss on rod and cone signaling and degeneration** *Hum Mol Genet* **31**:1035–1050

Jarvis MF, et al. (2007) **A-803467, a potent and selective Nav1.8 sodium channel blocker, attenuates neuropathic and inflammatory pain in the rat** *Proc Natl Acad Sci U S A* **104**:8520–8525

Jegla T, Simonson BT, Spafford JD (2024) **A broad survey of choanoflagellates revises the evolutionary history of the Shaker family of voltage-gated K(+) channels in animals** *Proc Natl Acad Sci U S A* **121**:e2407461121

Johnson B, Leek AN, Sole L, Maverick EE, Levine TP, Tamkun MM (2018) **Kv2 potassium channels form endoplasmic reticulum/plasma membrane junctions via interaction with VAPA and VAPB** *Proc Natl Acad Sci U S A* **115**:E7331–e7340

Jorge BS, Campbell CM, Miller AR, Rutter ED, Gurnett CA, Vanoye CG, George AL, Kearney JA (2011) **Voltage-gated potassium channel KCNV2 (Kv8.2) contributes to epilepsy susceptibility** *Proc Natl Acad Sci U S A* **108**:5443–5448

Kerscheneiner D, Stocker M (1999) **Heteromeric assembly of Kv2.1 with Kv9.3: effect on the state dependence of inactivation** *Biophys J* **77**:248–257

Kirmiz M, Palacio S, Thapa P, King AN, Sack JT, Trimmer JS (2018) **Remodeling neuronal ER-PM junctions is a conserved nonconducting function of Kv2 plasma membrane ion channels** *Mol Biol Cell* :mbcE18050337

Kobertz WR (2018) **Oddballs in the Shaker family: Kv2-related regulatory subunits** *J Gen Physiol* **150**:1599–1601

- Kramer JW, Post MA, Brown AM, Kirsch GE (1998) **Modulation of potassium channel gating by coexpression of Kv2.1 with regulatory Kv5.1 or Kv6.1 alpha-subunits** *Am J Physiol* **274**:C1501–1510
- Lee MC, Nahorski MS, Hockley JRF, Lu VB, Ison G, Pattison LA, Callejo G, Stouffer K, Fletcher E, Brown C, Drissi I, Wheeler D, Ernfors P, Menon D, Reimann F, Smith ESJ, Woods CG (2020a) **Human Labor Pain Is Influenced by the Voltage-Gated Potassium Channel K(V)6.4 Subunit** *Cell Rep* **32**:107941
- Lee MC, Nahorski MS, Hockley JRF, Lu VB, Ison G, Pattison LA, Callejo G, Stouffer K, Fletcher E, Brown C, Drissi I, Wheeler D, Ernfors P, Menon D, Reimann F, Smith ESJ, Woods CG (2020b) **Human labor pain is influenced by the voltage-gated potassium channel Kv 6.4 subunit** *Cell Rep* **32**:107941
- Li XN, Herrington J, Petrov A, Ge L, Eiermann G, Xiong Y, Jensen MV, Hohmeier HE, Newgard CB, Garcia ML, Wagner M, Zhang BB, Thornberry NA, Howard AD, Kaczorowski GJ, Zhou YP (2013) **The role of voltage-gated potassium channels Kv2.1 and Kv2.2 in the regulation of insulin and somatostatin release from pancreatic islets** *J Pharmacol Exp Ther* **344**:407–416
- Liu PW, Bean BP (2014) **Kv2 channel regulation of action potential repolarization and firing patterns in superior cervical ganglion neurons and hippocampal CA1 pyramidal neurons** *J Neurosci* **34**:4991–5002
- Liu Y, Beyer A, Aebersold R (2016) **On the Dependency of Cellular Protein Levels on mRNA Abundance** *Cell* **165**:535–550
- Lossin C, Wang DW, Rhodes TH, Vanoye CG, George AL (2002) **Molecular basis of an inherited epilepsy** *Neuron* **34**:877–884
- Maffie JK, Dvoretzkova E, Bougis PE, Martin-Eauclaire MF, Rudy B (2013) **Dipeptidyl-peptidase-like-proteins confer high sensitivity to the scorpion toxin AmmTX3 to Kv4-mediated A-type K⁺ channels** *J Physiol* **591**:2419–2427
- Malysz J, Petkov GV (2020) **Urinary bladder smooth muscle ion channels: expression, function, and regulation in health and disease** *Am J Physiol Renal Physiol* **319**:F257–f283
- Marino M, Misuri L, Brogioli D (2014) **A new open source software for the calculation of the liquid junction potential between two solutions according to the stationary Nernst-Planck equation** *arXiv* <https://doi.org/10.48550/arXiv.1403.3640>
- Marquis MJ, Sack JT (2022) **Mechanism of use-dependent Kv2 channel inhibition by RY785** *J Gen Physiol* **154**:e202112981
- Matsumoto C, O'Dwyer SC, Manning D, Hernandez-Hernandez G, Rhana P, Fong Z, Sato D, Clancy CE, Vierra NC, Trimmer JS, Fernando Santana L (2023) **The formation of K(V)2.1 macro-clusters is required for sex-specific differences in L-type Ca(V)1.2 clustering and function in arterial myocytes** *Commun Biol* **6**:1165
- Maverick EE, Leek AN, Tamkun MM (2021) **Kv2 channel-AMIGO β -subunit assembly modulates both channel function and cell adhesion molecule surface trafficking** *J Cell Sci* **134**
- Milescu M, Lee HC, Bae CH, Kim JI, Swartz KJ (2013) **Opening the shaker K⁺ channel with hanatoxin** *J Gen Physiol* **141**:203–216

- Milescu M, Bosmans F, Lee S, Alabi AA, Kim JI, Swartz KJ (2009) **Interactions between lipids and voltage sensor paddles detected with tarantula toxins** *Nat Struct Mol Biol* **16**:1080–1085
- Misonou H, Thompson SM, Cai X (2008) **Dynamic regulation of the Kv2.1 voltage-gated potassium channel during brain ischemia through neuroglial interaction** *J Neurosci* **28**:8529–8538
- Misonou H, Mohapatra DP, Menegola M, Trimmer JS (2005) **Calcium- and metabolic state-dependent modulation of the voltage-dependent Kv2.1 channel regulates neuronal excitability in response to ischemia** *J Neurosci* **25**:11184–11193
- Misonou H, Menegola M, Mohapatra DP, Guy LK, Park KS, Trimmer JS (2006) **Bidirectional activity-dependent regulation of neuronal ion channel phosphorylation** *J Neurosci* **26**:13505–13514
- Miyamae T, Hashimoto T, Abraham M, Kawabata R, Koshikizawa S, Bian Y, Nishihata Y, Kikuchi M, Ermentrout GB, Lewis DA, Gonzalez-Burgos G (2021) **Kcns3 deficiency disrupts Parvalbumin neuron physiology in mouse prefrontal cortex: Implications for the pathophysiology of schizophrenia** *Neurobiol Dis* **155**:105382
- Murakoshi H, Shi G, Scannevin RH, Trimmer JS (1997) **Phosphorylation of the Kv2.1 K⁺ channel alters voltage-dependent activation** *Mol Pharmacol* **52**:821–828
- Pathak D, Guan D, Foehring RC (2016) **Roles of specific Kv channel types in repolarization of the action potential in genetically identified subclasses of pyramidal neurons in mouse neocortex** *J Neurophysiol* **115**:2317–2329
- Peltola MA, Kuja-Panula J, Lauri SE, Taira T, Rauvala H (2011) **AMIGO is an auxiliary subunit of the Kv2.1 potassium channel** *EMBO Rep* **12**:1293–1299
- Pisupati A, Mickolajczyk KJ, Horton W, van Rossum DB, Anishkin A, Chintapalli SV, Li X, Chu-Luo J, Busey G, Hancock WO, Jegla T (2018) **The S6 gate in regulatory Kv6 subunits restricts heteromeric K⁺** *J Gen Physiol* **150**:1702–1721
- Plant LD, Dowdell EJ, Dementieva IS, Marks JD, Goldstein SA (2011) **SUMO modification of cell surface Kv2.1 potassium channels regulates the activity of rat hippocampal neurons** *J Gen Physiol* **137**:441–454
- Post MA, Kirsch GE, Brown AM (1996) **Kv2.1 and electrically silent Kv6.1 potassium channel subunits combine and express a novel current** *FEBS Lett* **399**:177–182
- Ray P, Torck A, Quigley L, Wangzhou A, Neiman M, Rao C, Lam T, Kim JY, Kim TH, Zhang MQ, Dussor G, Price TJ (2018) **Comparative transcriptome profiling of the human and mouse dorsal root ganglia: an RNA-seq-based resource for pain and sensory neuroscience research** *Pain* **159**:1325–1345
- Regnier G, Bocksteins E, Van de Vijver G, Snyders DJ, van Bogaert PP (2016) **The contribution of Kv2.2-mediated currents decreases during the postnatal development of mouse dorsal root ganglion neurons** *Physiol Rep* **4**:e12731
- Regnier G, Bocksteins E, Marei WF, Pintelon I, Timmermans JP, Leroy JLMR, Snyders DJ (2017) **Targeted deletion of the Kv6.4 subunit causes male sterility due to disturbed spermiogenesis** *Reprod Fertil Dev* **29**:1567–1575

- Renigunta V, Xhaferri N, Shaikh IG, Schlegel J, Bisen R, Sanvido I, Kalpachidou T, Kummer K, Oliver D, Leitner MG, Lindner M (2024) **A versatile functional interaction between electrically silent Kv subunits and Kv7 potassium channels** *Cell Mol Life Sci* **81**:301
- Richardson FC, Kaczmarek LK (2000) **Modification of delayed rectifier potassium currents by the Kv9.1 potassium channel subunit** *Hear Res* **147**:21–30
- Salinas M, de Weille J, Guillemare E, Lazdunski M, Hugnot JP (1997a) **Modes of regulation of shab K⁺ channel activity by the Kv8.1 subunit** *J Biol Chem* **272**:8774–8780
- Salinas M, Duprat F, Heurteaux C, Hugnot JP, Lazdunski M (1997b) **New modulatory alpha subunits for mammalian Shab K⁺ channels** *The Journal of biological chemistry* **272**:24371–24379
- Sano Y, Mochizuki S, Miyake A, Kitada C, Inamura K, Yokoi H, Nozawa K, Matsushime H, Furuichi K (2002) **Molecular cloning and characterization of Kv6.3, a novel modulatory subunit for voltage-gated K(+) channel Kv2.1** *FEBS Lett* **512**:230–234
- Sapio MR, Vazquez FA, Loydpierson AJ, Maric D, Kim JJ, LaPaglia DM, Puhl HL, Lu VB, Ikeda SR, Mannes AJ, Iadarola MJ (2020) **Comparative Analysis of Dorsal Root, Nodose and Sympathetic Ganglia for the Development of New Analgesics** *Front Neurosci* **14**:615362
- Sepela RJ, Stewart RG, Valencia LA, Thapa P, Wang Z, Cohen BE, Sack JT (2022) **The AMIGO1 adhesion protein activates Kv2.1 voltage sensors** *Biophys J* **121**:1395–1416
- Simonson BT, Jiang Z, Ryan JF, Jegla T (2025) **Ctenophores and parahoxozoans independently evolved functionally diverse voltage-gated K⁺ channels** *J Gen Physiol* **157**
- Stas JI, Bocksteins E, Labro AJ, Snyders DJ (2015) **Modulation of Closed-State Inactivation in Kv2.1/Kv6.4 Heterotetramers as Mechanism for 4-AP Induced Potentiation** *PLoS One* **10**:e0141349
- Stewart RG, Camacena M, Copits BA, Sack JT (2024) **Distinct cellular expression and subcellular localization of Kv2 voltage-gated K(+) channel subtypes in dorsal root ganglion neurons conserved between mice and humans** *The Journal of comparative neurology* **532**:e25575
- Thapa P, Stewart R, Sepela RJ, Vivas O, Parajuli LK, Lillya M, Fletcher-Taylor S, Cohen BE, Zito K, Sack JT (2021) **EVAP: A two-photon imaging tool to study conformational changes in endogenous Kv2 channels in live tissues** *J Gen Physiol* **153**:e202012858
- Thorneloe KS, Nelson MT (2003) **Properties and molecular basis of the mouse urinary bladder voltage-gated K⁺ current** *J Physiol* **549**:65–74
- Tilley DC, Angueyra JM, Eum KS, Kim H, Chao LH, Peng AW, Sack JT (2019) **The tarantula toxin GxTx detains K⁺** *J Gen Physiol* **151**:292–315
- Tilley DC, Eum KS, Fletcher-Taylor S, Austin DC, Dupré C, Patrón LA, Garcia RL, Lam K, Yarov-Yarovoy V, Cohen BE, Sack JT (2014) **Chemoselective tarantula toxins report voltage activation of wild-type ion channels in live cells** *Proc Natl Acad Sci U S A* **111**:E4789–E4796
- Toledo-Aral JJ, Moss BL, He ZJ, Koszowski AG, Whisenand T, Levinson SR, Wolf JJ, Silos-Santiago I, Halegoua S, Mandel G (1997) **Identification of PN1, a predominant voltage-dependent**

sodium channel expressed principally in peripheral neurons *Proc Natl Acad Sci U S A* **94**:1527–1532

Trapani JG, Korn SJ (2003) **Control of ion channel expression for patch clamp recordings using an inducible expression system in mammalian cell lines** *BMC Neurosci* **4**:15

Trimmer JS (1993) **Expression of Kv2.1 delayed rectifier K⁺ channel isoforms in the developing rat brain** *FEBS Lett* **324**:205–210

Tsantoulas C, Denk F, Signore M, Nassar MA, Futai K, McMahon SB (2018) **Mice lacking Kcns1 in peripheral neurons show increased basal and neuropathic pain sensitivity** *Pain* **159**:1641–1651

Tsantoulas C, Zhu L, Shaifta Y, Grist J, Ward JP, Raouf R, Michael GJ, McMahon SB (2012) **Sensory neuron downregulation of the Kv9.1 potassium channel subunit mediates neuropathic pain following nerve injury** *J Neurosci* **32**:17502–17513

Valtcheva MV, Copits BA, Davidson S, Sheahan TD, Pullen MY, McCall JG, Dikranian K, Gereau RWt (2016) **Surgical extraction of human dorsal root ganglia from organ donors and preparation of primary sensory neuron cultures** *Nat Protoc* **11**:1877–1888

Vierra NC, O'Dwyer SC, Matsumoto C, Santana LF, Trimmer JS (2021) **Regulation of neuronal excitation-transcription coupling by Kv2.1-induced clustering of somatic L-type Ca(2⁺) channels at ER-PM junctions** *Proc Natl Acad Sci U S A* **118**:e2110094118

Walker JR, Novick PA, Parsons WH, McGregor M, Zablocki J, Pande VS, Du Bois J (2012) **Marked difference in saxitoxin and tetrodotoxin affinity for the human nociceptive voltage-gated sodium channel (Nav1.7) [corrected]** *Proc Natl Acad Sci U S A* **109**:18102–18107

Wu H, Cowing JA, Michaelides M, Wilkie SE, Jeffery G, Jenkins SA, Mester V, Bird AC, Robson AG, Holder GE, Moore AT, Hunt DM, Webster AR (2006) **Mutations in the gene KCNV2 encoding a voltage-gated potassium channel subunit cause "cone dystrophy with supernormal rod electroretinogram" in humans** *Am J Hum Genet* **79**:574–579

Zheng Y, Liu P, Bai L, Trimmer JS, Bean BP, Ginty DD (2019) **Deep sequencing of somatosensory neurons reveals molecular determinants of intrinsic physiological properties** *Neuron* **103**:598–616

Zhong XZ, Abd-Elrahman KS, Liao CH, El-Yazbi AF, Walsh EJ, Walsh MP, Cole WC (2010) **Stromatoxin-sensitive, heteromultimeric Kv2.1/Kv9.3 channels contribute to myogenic control of cerebral arterial diameter** *J Physiol* **588**:4519–4537

Zylka MJ, Rice FL, Anderson DJ (2005) **Topographically distinct epidermal nociceptive circuits revealed by axonal tracers targeted to Mrgprd** *Neuron* **45**:17–25

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Reviewer #1 (Public review):**Summary:**

Kv2 subfamily potassium channels contribute to delayed rectifier currents in virtually all mammalian neurons and are encoded by two distinct types of subunits: Kv2 alpha subunits that have the capacity to form homomeric channels (Kv2.1 and Kv2.2), and KvS or silent subunits (Kv5,6,8,9) that can assemble with Kv2.1 or Kv2.2 to form heteromeric channels with novel biophysical properties. Many neurons express both types of subunits and therefore have the capacity to make both homomeric Kv2 channels and heteromeric Kv2/KvS channels. Determining the contributions of each of these channel types to native potassium currents has been very difficult because the differences in biophysical properties are modest and there are no Kv2/KvS-specific pharmacological tools. The authors set out to design a strategy to separate Kv2 and Kv2/KvS currents in native neurons based on their observation that Kv2/KvS channels have little sensitivity to the Kv2 pore blocker RY785 but are blocked by the Kv2 VSD blocker GxTx. They clearly demonstrate that Kv2/KvS currents can be differentiated from Kv2 currents in native neurons using a two-step strategy to first selectively block Kv2 with RY785, and then block both with GxTx. The manuscript is beautifully written; takes a very complex problem and strategy and breaks it down so both channel experts and the broad neuroscience community can understand it.

Strengths:

The compounds the authors use are highly selective and unlikely to have significant confounding cross-reactivity to other channel types. The authors provide strong evidence that all Kv2/KvS channels are resistant to RY785. This is a strength of the strategy - it can likely identify Kv2/KvS channels containing any of the 10 mammalian KvS subunits and thus be used as a general reagent on all types of neurons. The limitation then of course is that it can't differentiate the subtypes, but at this stage, the field really just needs to know how much Kv2/KvS channels contribute to native currents and this strategy provides a sound way to do so.

Weaknesses:

The authors are very clear about the limitations of their strategy, the most important of which is that they can't differentiate different subunit combinations of Kv2/KvS heteromers. This study is meant to be a start to understanding the roles of Kv2/KvS channels in vivo. As such, this is a minor weakness, far outweighed by the potential of the strategy to move the field through a roadblock that has existed since its inception.

The study accomplishes exactly what it set out to do: provide a means to determine the relative contributions of homomeric Kv2 and heteromeric Kv2/KvS channels to native delayed rectifier K⁺ currents in neurons. It also does a fabulous job laying out the case for why this is important to do.

Comments on revisions:

I liked this manuscript the first time and thought it was a great attempt to address a difficult problem, made more difficult by confusing background literature and conventions. The authors have kept all the strong points I liked from the first round and made it even stronger with their thoughtful and substantive responses to reviews. My first review was strongly supportive, and my initial short assessment/public review was written with the assumption that they would be public and the paper would be published essentially in its original form. All those points still apply so I am going to leave the initial reviews as is. The paper is a pleasure to read and a nice contribution to the field.

Reviewer #2 (Public review):

The authors used combined blockers/modulators to dissect the potassium currents mediated by inter-subunit heteromeric Kv channels. The method is robust given that the researchers know their limitations. Nevertheless, the authors elegantly tested their hypotheses, making this manuscript friendly to read despite the depth of all aspects they dealt with.

The quality of the data presented will positively impact the science involved in the study heteromeric channels, with clear developments in the field. Finally, the approach presented may unlock new studies related to these channels.

Comments on revisions:

The authors clarified all my points and beyond, specifically by adding some computational work that will also contribute to the subfield of heteromeric Kv channels.

<https://doi.org/10.7554/eLife.99410.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Kv2 subfamily potassium channels contribute to delayed rectifier currents in virtually all mammalian neurons and are encoded by two distinct types of subunits: Kv2 alpha subunits that have the capacity to form homomeric channels (Kv2.1 and Kv2.2), and KvS or silent subunits (Kv5,6,8,9) that can assemble with Kv2.1 or Kv2.2 to form heteromeric channels with novel biophysical properties. Many neurons express both types of subunits and therefore have the capacity to make both homomeric Kv2 channels and heteromeric Kv2/KvS channels. Determining the contributions of each of these channel types to native potassium currents has been very difficult because the differences in biophysical properties are modest and there are no Kv2/KvS-specific pharmacological tools. The authors set out to design a strategy to separate Kv2 and Kv2/KvS currents in native neurons based on their observation that Kv2/KvS channels have little sensitivity to the Kv2 pore blocker RY785 but are blocked by the Kv2 VSD blocker GxTx. They clearly demonstrate that Kv2/KvS currents can be differentiated from Kv2 currents in native neurons using a two-step strategy to first selectively block Kv2 with RY785, and then block both with GxTx. The manuscript is beautifully written; takes a very complex problem and strategy and breaks it down so both channel experts and the broad neuroscience community can understand it.

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is that it can't differentiate the subtypes, but at this stage, the field really just needs to know how much Kv2/KvS channels contribute to native currents and this strategy provides a sound way to do so.

Weaknesses:

The authors are very clear about the limitations of their strategy, the most important of which is that they can't differentiate different subunit combinations of Kv2/KvS heteromers. This study is meant to be a start to understanding the roles of Kv2/KvS channels in vivo. As such, this is a minor weakness, far outweighed by the potential of the strategy to move the field through a roadblock that has existed since its inception.

The study accomplishes exactly what it set out to do: provide a means to determine the relative contributions of homomeric Kv2 and heteromeric Kv2/KvS channels to native delayed rectifier K⁺ currents in neurons. It also does a fabulous job laying out the case for why this is important to do.

Reviewer #2 (Public Review):

Summary:

Silent Kv subunits and the channels containing these Kv subunits (Kv2/KvS heteromers) are in the process of discovery. It is believed that these channels fine-tune the voltage-activated K⁺ currents that repolarize the membrane potential during action potentials, with a direct effect on cell excitability, mostly by determining action potentials firing frequency.

Strengths:

What makes silent Kv subunits even more important is that, by being expressed in specific tissues and cell types, different silent Kv subunits may have the ability to fine-tune the delayed rectifying voltage-activated K⁺ currents that are one of the currents that crucially determine cell excitability in these cells. The present manuscript introduces a pharmacological method to dissect the voltage-activated K⁺ currents mediated by Kv2/KvS heteromers as a means of starting to unveil their importance, together with Kv2-only channels, to the cells where they are expressed.

Weaknesses:

While the method is effective in quantifying these currents in any isolated cell under an electric voltage clamp, it is ineffective as a modulating maneuver to perhaps address these currents in an in vivo experimental setting. This is an important point but is not a claim made by the authors.

We agree. We have now stated in the introduction that this study does not address the roles of Kv2/KvS currents in an in vivo setting.

Manuscript revisions:

While this study does not address the impact of GxTX or RY785 on action potentials or in vivo, the distinct pharmacology of Kv2/KvS heteromers presented here suggests that KvS conductances could be targeted to selectively modulate discrete subsets of cell types.

There are other caveats with the methods and data:

(i) The need for a 'cocktail' of blockers to supposedly isolate Kv2 homomers and Kv2/KvS heteromers' currents from others may introduce errors in the quantification Kv2/KvS heteromers-mediated K⁺ currents and that is due to possible blockers off targets.

We now point out that is possible that off target effects of blockers may introduce errors, include references that identify the selectivity of the blockers used in the cocktail, and specifically note that 4-aminopyridine in the cocktail is expected to block 2% of Kv2 homomers yet have a lesser impact Kv2/KvS heteromers. Additionally, to test whether the KvS isolation strategy requires the cocktail in neurons, we performed new experiments on a different subclass of nociceptors without the blocker cocktail and identified a substantial KvS-like component (new Fig 7 Supplement 3).

Manuscript revisions:

“After whole-cell voltage clamp was established, non-Kv2/KvS conductances were suppressed by changing to an external solution containing a cocktail of inhibitors: 100 nM alpha-dendrotoxin (Alomone) to block Kv1 (Harvey and Robertson, 2004), 3 μ M AmmTX3 (Alomone) to block Kv4 (Maffie et al., 2013; Pathak et al., 2016), 100 μ M 4-aminopyridine to block Kv3 (Coetzee et al., 1999; Gutman et al., 2005), 1 μ M TTX to block TTX sensitive Nav channels, and 10 μ M A803467 (Tocris) to block Nav1.8 (Jarvis et al., 2007). It is possible that off target effects of blockers may introduce errors in the quantification Kv2/KvS heteromer-mediated K⁺ currents. For example, 4-aminopyridine is expected to block a small fraction, 2%, of Kv2 homomers and have a lesser impact on Kv2/KvS heteromers (Post et al., 1996; Thorneloe and Nelson, 2003; Stas et al., 2015) which could result in a slight overestimation of the ratio of Kv2/KvS heteromers to Kv2 homomers.”

“We also tested the other major mouse C-fiber nociceptor population, peptidergic nociceptors, to determine if this subpopulation also has conductances resistant to RY785 yet sensitive to GxTX. We voltage clamped DRG neurons from a *CGRP^{GFP}* mouse line that expresses GFP in peptidergic nociceptors (Gong et al., 2003). Deep sequencing has identified mRNA transcripts for Kv6.2, Kv6.3, Kv8.1 and Kv9.3 present in GFP⁺ neurons, an overlapping but distinct set of KvS subunits from the *Mrgprd^{GFP}* non-peptidergic population (Zheng et al., 2019). In GFP⁺ neurons from *CGRP^{GFP}* mice, we found that a fraction of outward current was inhibited by 1 μ M RY785 and additional current inhibited by 100 nM GxTX (Fig 7 Supplement 3 A-C). In these experiments, $58 \pm 2\%$ (mean $\pm$ SEM) was KvS-like (Fig 7 Supplement 3 D) identifying that KvSlike conductances are present in these peptidergic nociceptors. For *CGRP^{GFP}* neurons we did not include the Kv1, Kv3, Kv4, Nav and Cav channel inhibitor cocktail used for other neuron experiments, indicating that the cocktail of inhibitors is not required to identify KvS-like conductances.”

(ii) During the electrophysiology experiments, the authors use a holding potential that is not as negative as it is needed for the recording of the full population of the Kv2/KvS channels. Depolarized holding potentials lead to a certain level of inactivation of the channels, that vary according to the KvS involved/present in that specific population of channels. As a reminder, some KvS promote inactivation and others prevent inactivation. Therefore, the data must be interpreted as such.

We agree. We now point out that the physiological holding potentials used are insufficiently negative to relieve inactivation from all Kv2/KvS heteromeric channels. We also note that the ratio of Kv2-like to KvS-like conductance is expected to vary with voltage protocols.

Manuscript revisions:

“Neurons were held at a membrane potential of -74 mV to mimic a physiological resting potential. KvS subunits can profoundly shift the voltage-inactivation relation (Salinas et al., 1997a; Kramer et al., 1998; Kerschensteiner and Stocker, 1999) and this potential is likely insufficiently negative to relieve inactivation from all Kv2/KvS heteromeric channels. Also, the activation membrane potential is close to the half-maximal point of Kv2/KvS

conductances. Thus the ratio of Kv2-like to KvS-like conductance is expected to vary with voltage protocols.”

(iii) The analysis of conductance activation by using tail currents is only accurate when dealing with non-inactivating conductances. Also, in dealing with a heterogenous population of Kv2/KvS heteromers, heterogenous K⁺ conductance deactivation kinetics is a must. Indeed, different KvS may significantly relate to different deactivation kinetics as well.

We now discuss that the bi-exponential fit of tail currents is likely inadequate to capture the deactivation kinetics of all underlying components of a heterogenous population of Kv2/KvS heteromers.

Manuscript revisions:

“We note that the analysis of conductance activation by using tail currents is only accurate when dealing with non-inactivating conductances. We expect that inactivation of Kv2/KvS conductances during the 200 ms pre-pulse is minimal (Salinas et al., 1997a; Kramer et al., 1998; Kerscheneiner and Stocker, 1999) and did not notice inactivation during the activation pulse. Also, deactivation kinetics can vary in a heterogenous population of Kv2/KvS heteromers. While analysis of tail currents could skew the quantification of total Kv2 like and KvS-like conductances, our data supports that mouse nociceptors and human neurons have tail currents that are resistant to RY785 and sensitive to GxTX consistent with the presence of Kv2/KvS heteromers.”

(iv) Silent Kv subunits may be retained in the ER, in heterologous systems like CHO cells. This aspect may subestimate their expression in these systems. Nevertheless, the authors show similar data in CHO cells and in primary neurons.

We agree. We now note that in heterologous systems, including CHO cells, transfection of KvS subunits can result in KvS subunits that are retained intracellularly.

Manuscript revisions:

“While a fraction of KvS subunits appear to be retained intracellularly, immunofluorescence for Kv5.1, Kv9.3 and Kv2.1 also appeared localized to the perimeter of transfected Kv2.1-CHO cells (Figure 1 Supplement).”

(v) The hallmark of silent Kv subunits is their effect on the time inactivation of K⁺ currents. As such, data should be shown throughout, preferably, from this perspective, but it was only done so in Figure 4G.

Indeed, effects on inactivation are a hallmark of KvS subunits. However, quantifying inactivation of Kv2/KvS channels requires steps to positive voltages for approximately 10 seconds. In neurons steps this long usually resulted in irreversible changes in leak currents/input resistance that degraded the accuracy of RY785/GxTX subtraction currents. Consequently, we did not acquire inactivation data in neurons, and we now explain in the manuscript why such data was not obtained.

Manuscript revisions:

“While changes in inactivation are prominent with KvS subunits, we did not investigate inactivation in neurons because the lengthy depolarizations required often resulted in irreversible leak current increases that degraded the accuracy of RY785/GxTX subtraction current quantification.”

(vi) Functional characterization of currents only, as suggested by the authors as a *bona fide* of Kv2 and Kv2/KvS currents, should not be solely trusted to classify the currents and their channel mediators.

We agree, and now state explicitly that functional characterization cannot be trusted to classify their channel mediators of conductances, and we try to be clear about this throughout the manuscript by using soft terms such as "KvS-like" when identity is uncertain.

Manuscript revisions:

"As functional characterization alone cannot be trusted to classify their channel mediators of conductances, we define conductances consistent with Kv2/KvS heteromers as 'KvS-like' and conductances consistent with Kv2 homomers as 'Kv2-like'."

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

There is not a lot to do here - this was a real pleasure to read and very easy to understand, as written. Here are a few minor things to consider:

(1) The naming of the KvS subunits has always been confusing - it is not clear that Kv5,6,8,9 are members of the Kv2 subfamily from the names. KvS does a good job of differentiating them by assembly phenotype and has been used a lot in the literature, but it doesn't solve the misconception of what subfamily they belong to. This might not matter so much for mammals, where all KvS channels are in the Kv2 subfamily, but it makes it impossible to extend the naming system to other animals where subunits requiring heteromeric assembly are common in most subfamilies. How about trying the name Kv2S? It would have continuity with KvS in the reader's mind, make it clear that they are Kv2 subfamily, and make a naming system that could be extended beyond vertebrates. This is not a problem the authors created - just a completely optional suggestion on how to solve it if so inclined.

We agree that naming conventions for these subunits are problematic, and agonized quite a bit about nomenclature. In the end we chose to stick with the precedent of KvS.

(2) Another naming issue they should definitely change is the use of "subfamily" for the different KvS subtypes (Kv5, Kv6, Kv8, and Kv9). This really creates confusion with the higher-order subfamilies that have a very clear functional definition: a subfamily of Kv genes is a group of related genes that have assembly compatibility. Those are Kv1, Kv2, Kv3 and Kv4. KvS genes are assembly compatible with Kv2, evolutionarily derived from the Kv2 lineage, and thus clearly a part of the Kv2 subfamily. Using a subfamily for the next lower level of the naming hierarchy confuses this. The authors should use different terms like sub-type or class or subgroups for the divisions within KvS.

Thank you. We have standardized to Kv2/KvS as a subfamily; Kv5, Kv6, Kv8, and Kv9 as subtypes; and individual proteins, e.g. Kv8.1, as subunits.

(3) When you discuss whether the KvS subunit directly disrupts Ry785 binding in the pore or works allosterically and you said you know which KvS residues point into the pore from models, I thought that maybe you could tell from a sequence alignment whether the KvS channels you didn't test look the same in the conduction pathway as the ones you did test. If so, you could mention that if the binding site is the pore, they should all be resistant. Alternatively, if one you didn't test looks fundamentally more similar to the

Kv2s in this region, then maybe it could be fingered as a possible exception that needs to be tested later.

Great ideas. We now assess sequence KvS variability near the proposed RY785 binding site in all KvS subunits. We generated structural models of RY785 docking to Kv2.1 and Kv2.1/Kv8.1 and found that residues near RY785 are different in all KvS subunits.

Manuscript revisions:

“We analyzed computational structural models of RY785 docked to a Kv2.1 homomer and a 3:1 Kv2.1:Kv8.1 heteromer (Fig 9) to gain structural insight into how KvS subunits might interfere with RY785 binding. We used Rosetta to dock RY785 to a cryo-EM structure of a Kv2.1 homomer in an apparently open state (Fernández-Mariño et al., 2023). The top-scoring docking pose has RY785 positioned below the selectivity filter and off-axis of the pore (Fig 9 A), similar to a stable pose observed in molecular dynamic simulations (Zhang et al., 2024). In this pose, RY785 contacts a collection of Kv2.1 residues that vary in every KvS subtype (Fig 9 B,D,E). Notably, RY785 bound similarly to a 3:1 model of Kv2.1/Kv8.1, in contact with the three Kv2.1 subunits, yet avoided the Kv8.1 subunit (Fig 9C). This is consistent with RY785 binding less well to Kv2.1/Kv8.1 heteromers, and also suggests that a 3:1 Kv2:KvS channel could retain a RY785 binding site when open.”

(4) Future suggestion or tip - not for this paper. Your data shows your isolation strategy works really well on Kv6 channels, and these are also the Kv2/KvS channels that have the most pronounced biophysical changes. Working on neurons that have a prominent Kv2/Kv6 component would really show how well the strategy outlined here works to describe the physiology of native neurons. The highest KvS expression I have seen in public data in a wellstudied cell type is Kv6.4 in spinal motor neurons.

Wonderful tip, thank you. We are indeed very interested in Kv6.4 in spinal motor neurons.

Reviewer #2 (Recommendations For The Authors):

The manuscript makes a good contribution to the identification of Kv2/KvS channels in primary cells. The pharmacological method proposed by the authors to dissect the currents in an experimental setting seems proper. Although meritorious in themselves, the findings are heavily phenomenological in the opinion of this reviewer. The manuscript should be improved with some level of mechanistic data and/or the demonstration of different levels of expression in different cell types.

Thank you for the suggestions. This manuscript now demonstrates strikingly higher levels of the KvS-like component of Kv2 currents in somatosensory (DRG nonpeptidergic and peptidergic nociceptor) versus autonomic (SCG) neuron types. The mechanistic question of what electrophysiological properties the KvS subunits are providing to the neuronal circuit is an exciting one that we are pursuing separately.

Manuscript revisions:

“While we found only RY785-sensitive Kv2-like conductances in SCG neurons, Kv2/KvS heteromer-like conductances were dominant in DRG neurons.”

At present, the manuscript says that the combination of RY785 and guangxitoxin-1E can be used to define Kv2/KvS-mediated K⁺ currents. Importantly, this method cannot be used in a way that one can functionally determine the function of Kv2/KvS channels, since it depends on the pre-blocking of Kv2-mediated K⁺ currents prior. In the opinion of this reviewer, this fact decreases the attention of a potential reader.

Indeed, our study is focused on revealing KvS heteromers by voltage clamp, and we now clarify in the introduction that we do not determine the function of Kv2/KvS channels in this study, so as not to lead the reader to expect studies of neuronal signaling.

However, the selective pharmacology we identify suggests RY785 application could reveal the function of Kv2 homomers, and for RY785-insensitive signaling, GxTX application could reveal the function of Kv2/KvS heteromers. We now mention these possible applications in the Discussion.

Manuscript revisions:

“While this study does not address the impact of GxTX or RY785 on action potentials or in vivo, the distinct pharmacology of Kv2/KvS heteromers presented here suggests that KvS conductances could be targeted to selectively modulate discrete subsets of cell types.”

Please find below suggestions for improving the manuscript:

(1) The term "Kv2/KvS heteromers" should be used throughout instead of variations such as "Kv2/KvS channels", "Kv2/KvS" and others. Standardization of the term to refer to heteromers would make the manuscript easier to read.

Thank you. We have standardized terms to consistently refer to Kv2/KvS heteromers.

(2) Confusing terms like KvS conductances, KvS-like conductances, KvS-like (RY785-resistant, GxTX-sensitive) currents, and KvS channels should be avoided because they disregard the current belief that KvS cannot form functional homomeric channels. The term KvS-containing channels, and Kv2/KvS channels, seem more accurate. Uniformization in this regard will also make the manuscript more easily readable.

Thank you. We have standardized terms to Kv2/KvS heteromers and KvS-containing channels when channel subunits are known and the use terms KvS-like and Kv2-like for functionally identified endogenous conductances with unknown channel subunits.

(3) Referring to KvS as a regulatory subunit is inaccurate. It is clear that KvS is part of, and it makes up the alpha pore. KvS therefore is a part of the conductive pathway and not a regulatory (suggesting accessory) subunit. KvS take part in selectivity filter (fully conserved), but they also make up an important part of the conducting pathway with non-conserved amino acid residues.

We felt it important to include the descriptor “regulatory” to connect our nomenclature with prior use of the descriptor in the literature, and now only use the term at the start of the introduction.

Manuscript revisions:

“A potential source of molecular diversity for Kv2 channels are a group of Kv2-related proteins which have been referred to as regulatory, silent, or KvS subunits.”

(4) The use of a cocktail of channel inhibitors may affect the quantification of Kv2/KvS heteromers-mediated K⁺ currents because they may interact with RY785 and/or GxTx or they may even interact with the sites for these two drugs on Kv2-containing channels.

This is an interesting point worth considering, thank you. We now alert readers to this possibility in the discussion when considering the limitations of our approach.

Manuscript revisions:

“Also, the cocktail of inhibitors used in most neuron experiments here could potentially alter RY785 or GxTX action against KvS/Kv2 channels.”

(5) The graphical representation of fractional blocking and other parameters (e.g., Fig 1D), is hard to read in these slim plots. In my opinion, tall bars would be more meaningfully visualized.

Thank you for pointing out that the graphs were hard to read, we have made the graph easier to read and added tall bars.

(6) Vehicle control for IHC and electrophysiology. Please state what is the vehicle used in the electrophysiology experiments.

Thank you. The composition of vehicle has now been stated in the methods.

Manuscript revisions:

“All RY785 solutions contained 0.1% DMSO. Vehicle control solutions also contained 0.1% DMSO but lacked RY785.”

“Sections were incubated in vehicle solution (4% milk, 0.2% triton diluted in PB) for 1 hr at RT.”

(7) The reference Trapani & Korn, 2003 (?) is not included in the list. This reference is important since it sets what are the Kv2.1-CHO cells. In this regard it is also important to mention, even better to address, the expressing qualities of this system in the face of a co-expression with a plasmid-based expression of silent Kv subunits. Are these two ways of expressing Kv subunits, meant to come together (or not) in heteromers, balanced? This question is critical here. Still, in regard to Kv2.1-CHO cells, it was not clear in the manuscript if the term "transfection" refers only to the plasmids used to temporarily induce the expression of silent Kv subunits and potentially Kv channels accessory subunits.

We now include the Trapani & Korn, 2003 reference (thank you for pointing out this accidental omission), and better explain expression methods. The benefit of the inducible Kv2.1 expression is control of Kv conductance densities which can otherwise become so large as to be refractory to voltage clamp. The beauty of the expression system is that cells recently transfected with KvS subunits can be induced to express just enough Kv2.1 to get a substantial but not clampoverwhelming RY785-resistant Kv2/KvS conductance. We also discuss that our expression methods are distinct from past studies. We stop short of comparing the expression systems, as this is beyond the scope of what we set out to study.

Manuscript revisions: See next response

(8) Kv2.1-CHO cells transfection procedures, induction, and validation are unclear. This validation is important here.

We have clarified transfection procedures, induction, and validation in the methods section.

Manuscript revisions:

“The CHO-K1 cell line transfected with a tetracycline-inducible rat Kv2.1 construct (Kv2.1-CHO) (Trapani and Korn, 2003) was cultured as described previously (Tilley et al., 2014).”

Transfections were achieved with Lipofectamine 3000 (Life Technologies, L3000001). 1 μ l Lipofectamine was diluted, mixed, and incubated in 25 μ l of Opti-MEM (Gibco, 31985062).”

“Concurrently, 0.5 μ g of KvS or AMIGO1 or Nav β 2, 0.5 μ g of pEGFP, 2 μ l of P3000 reagent and 25 μ l of Opti-MEM were mixed. DNA and Lipofectamine 3000 mixtures were mixed and incubated at room temperature for 15 min. This transfection cocktail was added to 1 ml of culture media in a 24 well cell culture dish containing Kv2.1-CHO cells and incubated at 37 $^{\circ}$ C in 5% CO₂ for 6 h before the media was replaced. Immediately after media was replaced, Kv2.1 expression was induced in Kv2.1-CHO cells with 1 μ g/ml minocycline (Enzo Life Sciences, ALX380-109-M050), prepared in 70% ethanol at 2 mg/ml. Voltage clamp recordings were performed 12-24 hours later. We note that the expression method of Kv2/KvS heteromers used here is distinct from previous studies which show that the KvS:Kv2 mRNA ratio can affect the expression of functional Kv2/KvS heteromers (Salinas et al., 1997b; Pisupati et al., 2018). We validated the functional Kv2/KvS heteromer expression using voltage clamp to establish distinct channel kinetics and the presence of RY785-resistant conductance in KvS-transfected cells and using immunohistochemistry to label apparent surface localization of KvS subunits (Figure 4, Figure 1 Supplement, Figure 1 and Figure 5).”

(9) It is important for readers to add some context to Kv2.1/Kv8.1 channels (and other Kv2/KvS heteromers) used to test the combination of RY785 and GxTx. In my opinion, this enriches the discussion.

Good idea. We have added context about each of the KvS subunits we test.

Manuscript revisions:

“To test the pharmacological response of KvS we began with Kv8.1, a subunit that creates heteromers with biophysical properties distinct from Kv2 homomers (Salinas et al., 1997a), and modulates motor neuron vulnerability to cell death (Huang et al., 2024).

Each of these KvS subunits create Kv2/KvS heteromers that have distinct biophysical properties (Kramer et al., 1998; Kerschensteiner and Stocker, 1999; Bocksteins et al., 2012). Kv5.1/Kv2.1 heteromers play an important role in controlling the excitability of mouse urinary bladder smooth muscle (Malysz and Petkov, 2020), mutations in Kv6.4 have been shown to influence human labor pain (Lee et al., 2020b), and deficiency of Kv9.3 disrupts parvalbumin interneuron physiology in mouse prefrontal cortex (Miyamae et al., 2021).”

(10) In general, the membrane potential used to activate Kv2 only channels and Kv2/KvS channels is too close to the activation $V_{1/2}$. In case the comparing curves are displaced in their relative voltage dependence and voltage sensitivity, using that range of membrane potential may introduce a crucial error in the estimation of the conductance's relative amplitudes.

We now note that the relative conductances of Kv2-only vs Kv2/KvS channels are expected to vary with voltage protocol, as KvS inclusion results in channels with altered voltage responses.

Manuscript revisions:

“...the activation membrane potential is close to the half-maximal point of Kv2/KvS conductances. Thus the ratio of Kv2-like to KvS-like conductance is expected to vary with voltage protocols.”

(11) The use of tail currents to estimate conductance is problematic if i) lack of current inactivation is not assured, and ii) if the different currents, with possible different deactivation kinetics at the used membrane potential (e.g., mV), are not assured. Why was the activation peak used at times, and at different elapsed times the tail currents

were used instead? These aspects of conductance's amplitude estimation methods should be well defined.

In CHO cells peak currents were analyzed because outward currents seem to offer the best signal/noise. In neurons, we restricted analysis to tail currents at elapsed times to minimize complications from non-Kv2 endogenous voltage-gated channels which deactivate more quickly. We have clarified this analysis in the methods section.

Manuscript revisions:

“In CHO cells peak currents were analyzed because outward currents seem to offer the best signal/noise. In neurons, we restricted analysis to tail currents at elapsed times to minimize complications from non-Kv2 endogenous voltage-gated channels which deactivate more quickly. In neurons, voltage gated currents remained in the toxin cocktail + RY785 and GxTX, that were sometimes unstable. To minimize complications from these currents, we restricted analysis of RY785 and GxTX subtraction experiments to tail currents at elapsed times to minimize complications from non-Kv2 endogenous voltage-gated channels which deactivate more quickly. We note that the analysis of conductance activation by using tail currents is only accurate when dealing with non-inactivating conductances. We expect that inactivation of Kv2/KvS conductances during the 200 ms pre-pulse is minimal (Salinas et al., 1997a; Kramer et al., 1998; Kerschensteiner and Stocker, 1999) and did not notice inactivation during the activation pulse. Also, deactivation kinetics can vary in a heterogenous population of Kv2/KvS heteromers. While analysis of tail currents could skew the quantification of total Kv2 like and KvS-like conductances, our data supports that mouse nociceptors and human neurons have tail currents that are resistant to RY785 and sensitive to GxTX consistent with the presence of Kv2/KvS heteromers.”

(12) Were the experiments including different conditions such as control, RY, and RY+GxTx done pair-wised? This could potentially better the statistics and strengthen the data and the conclusions drawn from them.

The control, RY, and RY+GxTX in neurons were done pairwise and the statistical tests performed for these experiments were pairwise tests. We have clarified this in the figure legends.

Manuscript revisions:

“Wilcoxon rank tests were paired, except the comparison of RY785 to vehicle which was unpaired.”

(13) The holding potential of the experiments, mostly -89 mV, may be biasing the estimation of Kv2 only channels vs. Kv2/KvS channels conductances. Figure 4I exemplifies this concern.

We agree. Figure 4I reveals that a holding potential of -89 mV vs -129 mV reduces conductance of Kv2.1/Kv8.1 heteromers vs Kv2.1 homomers in CHO cells by ~20%. We have now alerted readers that the ratio of Kv2 only channels vs. Kv2/KvS conductances can vary with holding voltage.

Manuscript revisions:

“Under these conditions, $58 \pm 3\%$ (mean $\pm$ SEM) of the delayed rectifier conductance was resistant to RY785 yet sensitive to GxTX (KvS-like) (Fig 7 F). We note that the ratio of KvS- to Kv2-like conductances is expected to vary with holding potential, as KvS subunits can change the degree and voltage-dependence of steady state inactivation (e.g. Fig 4I).”

(14) *It is possible that Figure 6A (control trace) and Figure 6C ("Kv2-like" trace) are the same, by mistake, since their noise pattern looks too similar.*

Indeed the noise pattern of the Figure 6A (control trace) and Figure 6C ("Kv2-like" trace) are related, as they have inputs from the same trace, with Figure 6C ("Kv2-like" trace) being a subtraction of Figure 6A (+RY trace) from Figure 6A (control trace).

(15) *For example, in Figure 7A, what is the identity of the current remaining after the RY+GxTx application? In Figure 7B, a supposed outlier in the group of data referring to "veh" in the right panel is what possibly is making this group different from +RY in the left panel ($p=0.02$, Wilcoxon rank test). I would recommend parametric tests only since the data is essentially quantitative.*

In Figure 7A, we do not know the identity of the current remaining after the RY+GxTX application, the kinetics of the residual current appeared distinct from the Kv2/KvS-like currents blocked by RY or GxTX, but we did not analyze these.

The data in Figure 7B, was indeed the positive outlier in the group of data referring to "veh" in the right panel and contributes to the p-value, but we saw no reason to exclude it. We have now replaced the representative trace in 7B with a non-outlier trace. We respectfully disagree with the suggestion to use parametric statistical tests as we do not know the distribution underlying the variance our data.

Manuscript revisions:

"Subsequent application of 100 nM GxTX decreased tail currents by $68 \pm 5\%$ (mean $\pm$ SEM) of their original amplitude before RY785. We do not know the identity of the outward current that remains in the cocktail of inhibitors + RY785 + GxTX."

(16) *Please state the importance of using nonpeptidergic neurons to study silent Kv5.1 and Kv9.1 subunits. RNA data may not necessarily work to probe function or protein abundance, which is crucial in heteromeric complexes.*

We have now more thoroughly explained our rationale for choosing the nonpeptidergic neurons.

RNA is not predictive of protein abundance, and we have not yet been successful in measuring KvS protein abundance in these neurons, so we've probed KvS abundance by assessing RY785 resistance.

Manuscript revisions:

"Mouse dorsal root ganglion (DRG) somatosensory neurons express Kv2 proteins (Stewart et al., 2024), have GxTX-sensitive conductances (Zheng et al., 2019), and express a variety of KvS transcripts (Bocksteins et al., 2009; Zheng et al., 2019), yet transcript abundance does not necessarily correlate with functional protein abundance. To record from a consistent subpopulation of mouse somatosensory neurons which has been shown to contain GxTX-sensitive currents and have abundant expression of KvS mRNA transcripts (Zheng et al., 2019), we used a *Mrgprd*^{GFP} transgenic mouse line which expresses GFP in nonpeptidergic nociceptors (Zylka et al., 2005; Zheng et al., 2019). Deep sequencing identified that mRNA transcripts for Kv5.1, Kv6.2, Kv6.3, and Kv9.1 are present in GFP+ neurons of this mouse line (Zheng et al., 2019) and we confirmed the presence of Kv5.1 and Kv9.1 transcripts in GFP+ neurons from *Mrgprd*^{GFP} mice using RNAscope (Fig 7 Supplement 1)."

(17) *In Figure 8B, were +RY data different from veh data? The figure shows no Wilcoxon (nonparametric) comparison and this is important to be stated. What conductance(s) is the vehicle solution blocking or promoting? What is RY dissolved in, DMSO? What is the DMSO final concentration?*

We now state that in Figure 8B, +RY amplitudes were not statistically different from veh data in this limited data set. However, the RY-subtraction currents always had Kv2-like biophysical properties, whereas vehicle-subtraction currents had variable properties precluding biophysical analysis for Fig 8D.

In Figure 8B, we do not know what conductance(s) the vehicle solution is affecting, we think the changes observed are likely merely time dependent or due to the solution exchange itself. RY stock is in DMSO. All recording solutions have 0.1% DMSO final concentration, this is now noted in methods.

Manuscript revisions:

“Unlike mouse neurons, we did not detect a significant difference in tail currents of RY785 versus vehicle controls. However, RY785-subtracted currents always had Kv2-like biophysical properties whereas vehicle-subtraction currents had variable properties that precluded the same biophysical analysis. Overall, these results show that human DRG neurons can produce endogenous voltage-gated currents with pharmacology and gating consistent with Kv2/KvS heteromeric channels.”

“All RY785 solutions contained 0.1% DMSO. Vehicle control solutions also contained 0.1% DMSO but lacked RY785.”

(18) *METHODS. The electrophysiology approach should be unified in all aspects as applicable and possible.*

We have unified the mouse dorsal root ganglion and mouse superior cervical ganglion methods sections. We have kept CHO cells and mouse/human neurons section separate because the methods were substantially different.

(19) *DISCUSSION. The discussion section spends half of its space trying to elaborate on possible blocking/inhibiting/modulating mechanisms for RY785. The present manuscript shows no data, at least not that I have noticed, that would evoke such discussion.*

We have shortened this section, and enhance the discussion with structural models (new Fig 9), and our functional data indicating perturbed RY785 interaction with Kv2.1/8.1.

Manuscript revisions:

“In this pose, RY785 contacts a collection of Kv2.1 residues that vary in every KvS subtype (Fig 9 B,D,E). Notably, RY785 bound similarly to a 3:1 model of Kv2.1/Kv8.1, in contact with the three Kv2.1 subunits, yet avoided the Kv8.1 subunit (Fig 9C). This is consistent with RY785 binding less well to Kv2.1/Kv8.1 heteromers, and also suggests that a 3:1 Kv2:KvS channel could retain a RY785 binding site when open. However, the RY785 resistance of Kv2/KvS heteromers may primarily arise from perturbed interactions with the constricted central cavity of closed channels. In homomeric Kv2.1, RY785 becomes trapped in closed channels and prevents their voltage sensors from fully activating, indicating that RY785 must interact differently with closed channels (Marquis and Sack, 2022). Here we found that Kv2.1/Kv8.1 current rapidly recovers following washout of RY785, suggesting that Kv2.1/Kv8.1 heteromers do not readily trap RY785 (Figure 2 Supplement). Overall, the structural modeling suggests

that KvS subunits sterically interfere with RY785 binding to the central cavity, while functional data suggest KvS subunits disrupt RY785 trapping in closed states.”

(20) DISCUSSION. Topics like ER retention and release upon certain conditions would be a better enrichment for the manuscript in my opinion.

ER retention of KvS subunits is indeed an important topic! However, we have opted not to delve into it here.

(21) DISCUSSION. Speculation about the binding site for RY on Kv2/KvS channels is also not touched by the data shown in the manuscript.

We have shortened this section of discussion, and now present this with structural models of RY785 docked to a Kv2.1 homomer and 3:1 Kv2.1: Kv8.1 heteromer (new Fig 9) to ground speculations. See manuscript changes noted in response to comment (19) above.

(22) DISCUSSION. An important reference is missing in regard to stoichiometry: Bocksteins et al., 2017. This work is the only one using a non-optical technique to add knowledge to that question.

Good point, and an excellent study we didn’t realize we’d not included before. We now include Bocksteins et al. 2017 as a reference in the Introduction.

(23) In my opinion, allosterism and orthosterism are concepts not yet useful for the discussion of RY binding sites without even a general piece of data.

We now include structural models of RY785 docked to a Kv2.1 homomer and 3:1 Kv2.1: Kv8.1 heteromer (new Fig 9) to ground blocking speculations. See manuscript changes noted in response to comment (19).

(24) The term "homogeneously susceptible" associated with a Hill slope close to 1 needs to be more elaborated.

Thank you, we have elaborated.

Manuscript revisions:

“Also, the degree of resistance to RY785 may vary if Kv2:KvS subunit stoichiometry varies. With high doses of RY785, we found that the concentration-response characteristics of Kv2.1/Kv8.1 in CHO cells revealed hallmarks of a homogenous channel population with a Hill slope close to 1 (Fig 2B). However, other KvS subunits might assemble in multiple stoichiometries and result in pharmacologically-distinct heteromer populations.”

(25) Stating the KvS are resistant to RY785 is not proper in my opinion. This opinion relates to the fact that the RY binding site in the channels is certainly not restricted to a binding site residing only on the Kv subunit.

Good point. We have now changed phrasing to convey that KvS subunits are a component of a heteromer that imbues RY785 resistance.

Manuscript revisions:

“These results show that voltage-gated outward currents in cells transfected with members from each KvS subtype have decreased sensitivity to RY785 but remain sensitive to GxTX. While we did not test every KvS subunit, the ubiquitous resistance suggests that all KvS

subunits may provide resistance to 1 μ M RY785 yet remain sensitive to GxTX, and that RY785 resistance is a hallmark of KvS-containing channels.”

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